FACULTY OF COMPUTER APPLICATIONS

Marwadi University
Marwadi Chandarana Group

NAAC A+

Faculty of Computer Applications

BCA Sem-1 (AY-2026-27)

Lab Book

Sub Code : 05BC3103

(Programming Practices : 1)

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GitHub Link | https://github.com/tirthbhimani13-droid/bca-1-pp-1--92600527009-tirth-bhimani

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INDEX

Unit : 1	
Program No	**Definition**
1	Print “Hello World”.
2	Takes 2 Values from user and Perform AllArithmetic Operations.
3	Print numbers from 1 to 20.
4	Input 3 values from user and find out its Average.
5	Calculate the Square and Cube of a number.
6	Interchange the value of two numbers.
7	Find largest out of three numbers
8	Find out Fibonacci series up to given n values
9	Find out Number is Odd or Even
10	Find out sum of digits of a given number.
	UNIT-2
1	PRINT”Hello world”
2	Write a Program that takes 2 Values from user and Perform All Arithmetic Operation.
3	Write a program that takes 3 values for PrincipleAmount, Rate of Interest and No. of Years and find out Simple Interest.

4	Write a program that input value from user and find out Area of Circle.
5	Write a program that input two values from user and find out Area of Rectangle.
6	Write a program that input 3 values from user and find out its Average.
7	Write a program to Calculate the Square and Cube of a number.
8	Write a program Interchange the value of two numbers using third variable.
9	Write a program Interchange the value of two numbers without using third variable.
10	Write a program that input quantity, price, discount and display final amount.

(1) Enter N elements and find total and average of them.

```
#include<stdio.h>
#include<conio.h>

void main()
{
   int num[50],i,N,sum=0;
float avg;

   printf("\n Enter the total elements N : ");
scanf("%d",&N);

   for(i=0;i<N;i++)
   {
      printf("\n Enter value : %d ",i+1);
scanf("%d",&num[i]);
   }
   for(i=0;i<N;i++)
   {
     sum = sum + num[i];
```

```
    }
    avg = sum / N;

    printf("\n Sum of all values is : %d",sum);
printf("\n Average of all values is : %.2f",avg);

}
```

Output :

```
 Enter the total elements N : 3

 Enter value : 1 12

 Enter value : 2 34

 Enter value : 3 54

 Sum of all values is : 100
 Average of all values is : 33.00
Process returned 34 (0x22)   execution time : 14.015 s
Press any key to continue.
```

1. Print “Hello World”.

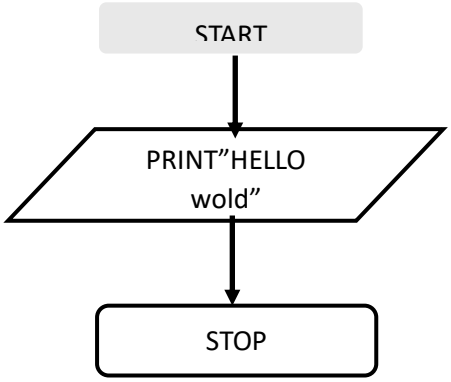
Algoritham :

Step 1 : START

Step 2 : PRINT “Hello World”

Step 3 : STOP

Flowchart :



Code :

```
#include <stdio.h>
#include <conio.h>

void main()
{
    clrscr();

    printf("hello");

    getch();

}
```

Output :



2. Takes 2 Values from user and Perform AllArithmetic Operations

Algorithm:

Step 1 : START

Step 2 : DECLARE variabales x=0.0, y=0.0,a, s, m, d

Step 3 : INPUT x,y

Step 4 : a= x + y

Step 5 : s= x – y

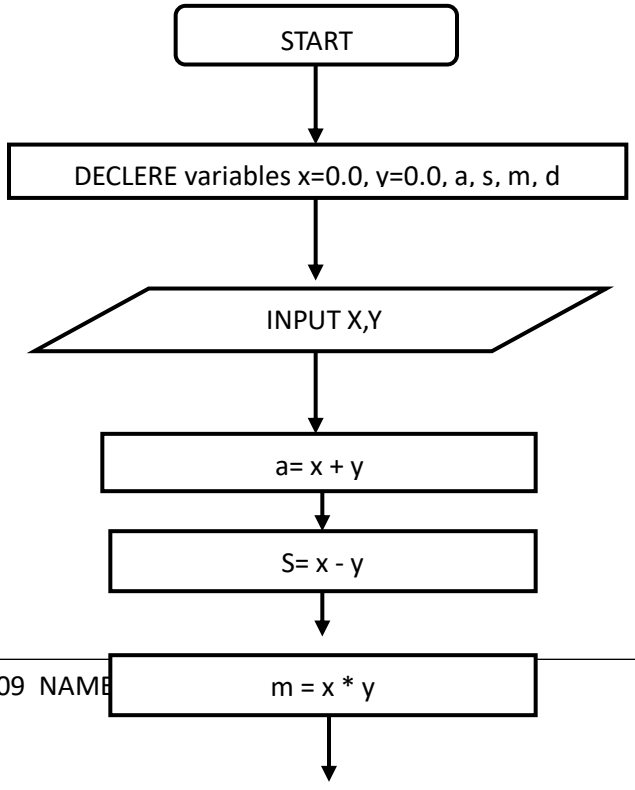
Step 6 : m= x * y

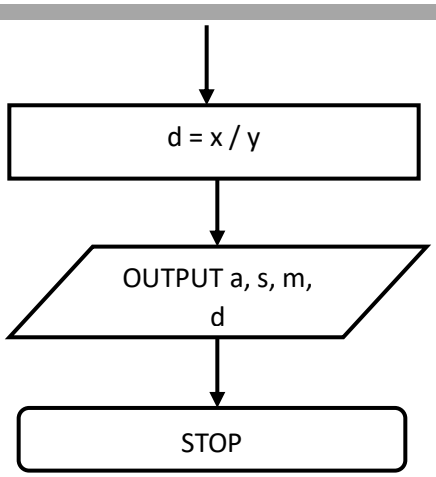
Step 7 : d = x / y

Step 8 : OUTPUT a, s, m, d

Step 9 : STOP

Floechart:





Code:

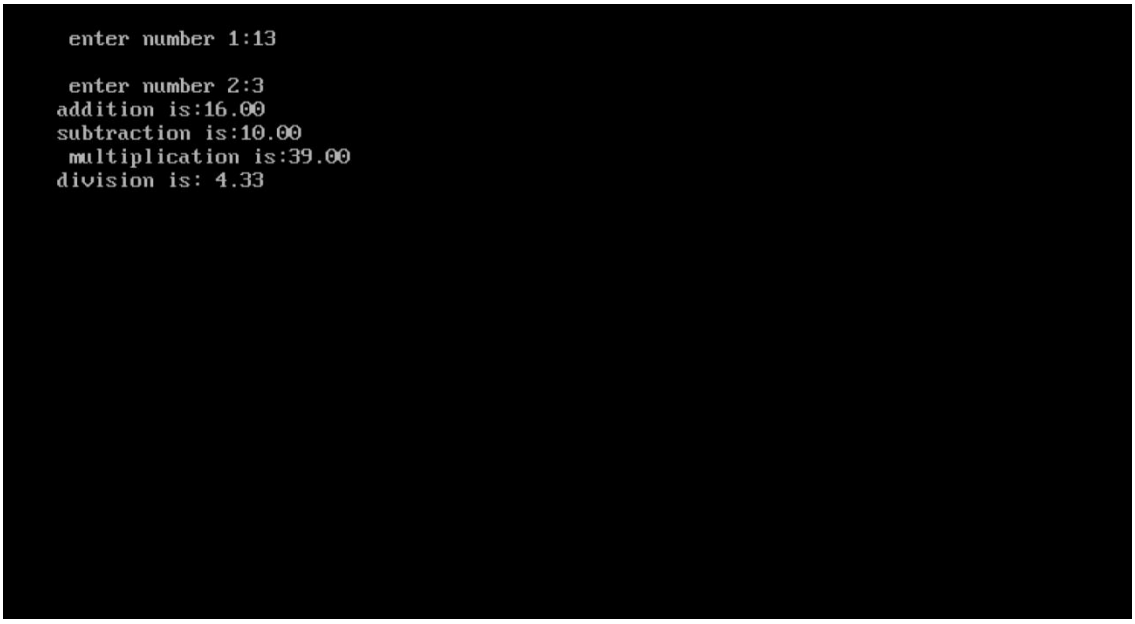
```
#include<stdio.h>
#include<conio.h>

void main()
{
   float x,y;
   clrscr();
   printf("\n Enter Number 1: ");
   scanf("%f",&x);
   printf("\n Enter Number 2: ");
   scanf("%f",&y);

   printf("Addition Is: %.2f \n",x+y);
   printf("Subtraction Is: %.2f \n",x-y);
   printf("Multiplication Is: %.2f \n",x*y);
   printf("Division Is: %.2f \n",x/y);
   getch();
```

}

Output :


enter number 1:13
enter number 2:3
addition is:16.00
subtraction is:10.00
multiplication is:39.00
division is: 4.33

4. Input 3 values from user and find out its Average.

Algorithm :

STOP Step 1 : START

Step 2 : DECLARE variabales A, B, C

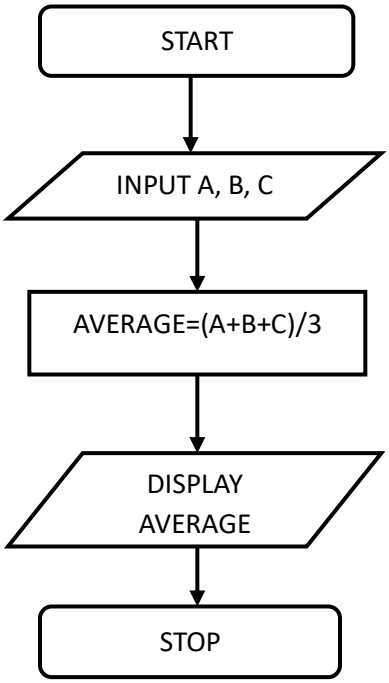
Step 3 : CALCULATE SUM = A+B+C

Step 4 : CALCULATE AVERAGE SUM/3

Step 5 : PRINT AVERAGE

Step 6 : STOP

Flowchart:



Code:

```
#include <stdio.h>
#include <conio.h>

void main() {
   float  A, B, C, average;
   clrscr();

   printf("Enter first number: ");
   scanf("%f", &A);

   printf("Enter second number: ");
   scanf("%f", &B);

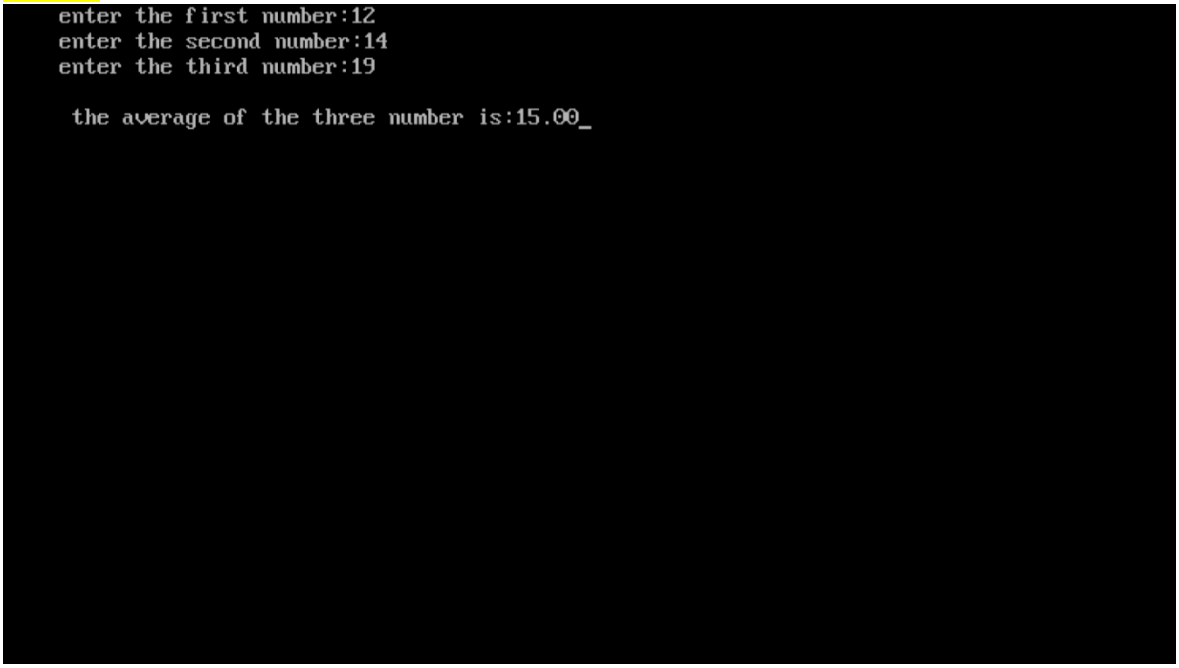
   printf("Enter third number: ");
   scanf("%f", &C);

   average = (A + B + C ) / 3.0;

   printf("\nThe Average is: %.2f", average);

   getch();
}
```

Output:


enter the first number:12
enter the second number:14
enter the third number:19
the average of the three number is:15.00_

5. Calculate the Square and Cube of a number.

Algorithm:

Step 1 : START

Step 2 : DECLARE variabales N

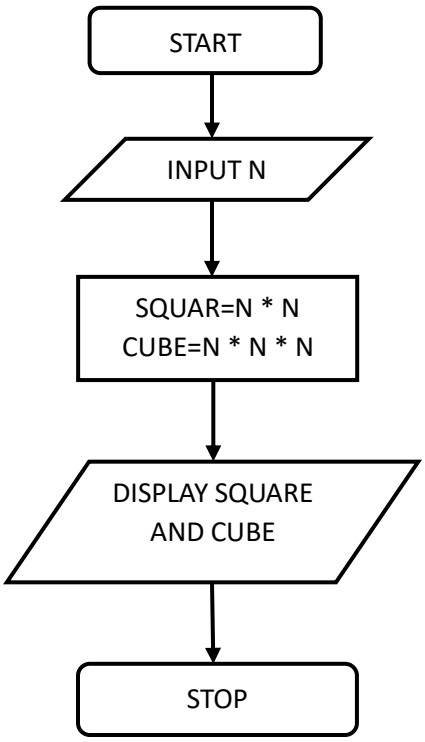
Step 3 : CALCULATE SQUARE = N * N

Step 4 : CALCULATE CUBE = N * N * N

Step 5 : PRINT ASQUARE AND CUBE

Step 6 : STOP

Flowchart:



Code :

```
#include <stdio.h>

int main() {
   int N, square, cube;

   printf("Enter an integer: ");
   scanf("%d", &N);

   square = N * N;
   cube = square * N;

   printf("Square of %d = %d\n", N, square);
   printf("Cube of %d = %d\n", N, cube);

   getch();
}
```

Output:



6. Interchange the value of two numbers

Algorithm:

Step 1: Start.

Step 2: Read two numbers into variables A and B.

Step 3: Copy value of A to temporary variable temp.

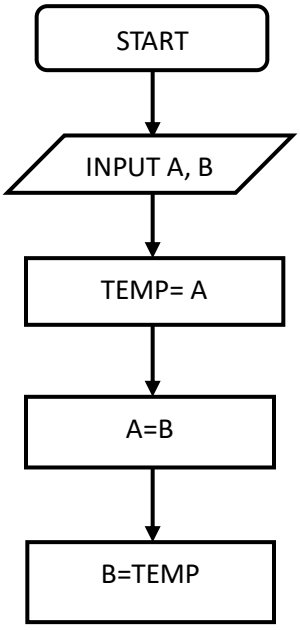
Step 4: Copy value of B to A.

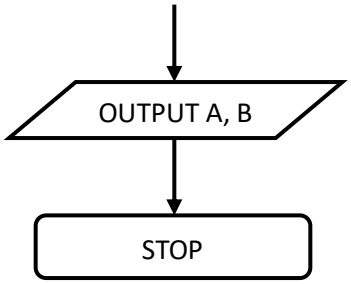
Step 5: Copy value of temp to B.

Step 6: Print values of A and B.

Step 7: Stop.

Flowchart:




OUTPUT A, B
STOP

Code:

```
#include <stdio.h>
#include <conio.h>

void main() {
   int A, B, temp;

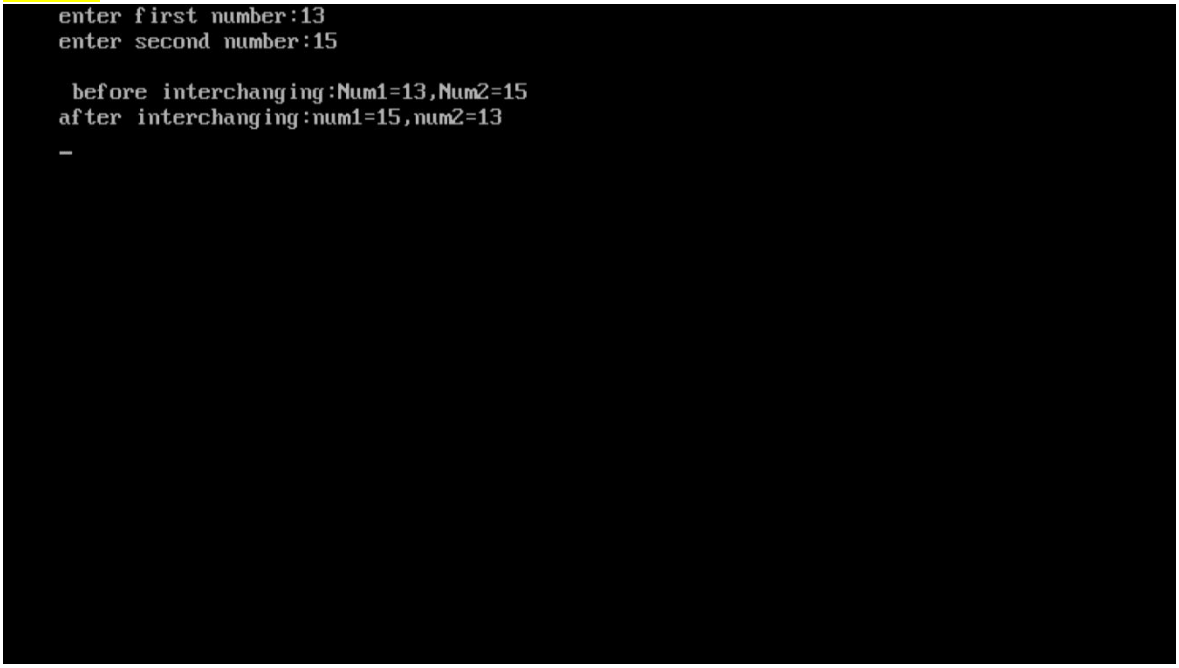
   clrscr();
   printf("Enter first number (A): ");
   scanf("%d", &A);
   printf("Enter second number (B): ");
   scanf("%d", &B);

   printf("\nBefore swapping: A = %d, B = %d\n", A, B);

   temp = A;
   A = B;
   B = temp;

   printf("After swapping: A = %d, B = %d\n", A, B);
   getch();
}
```

Output:



9. Find out Number is Odd or Even

Algorithm:

Step 1: Start

Step 2: Read a number & Declare variable N

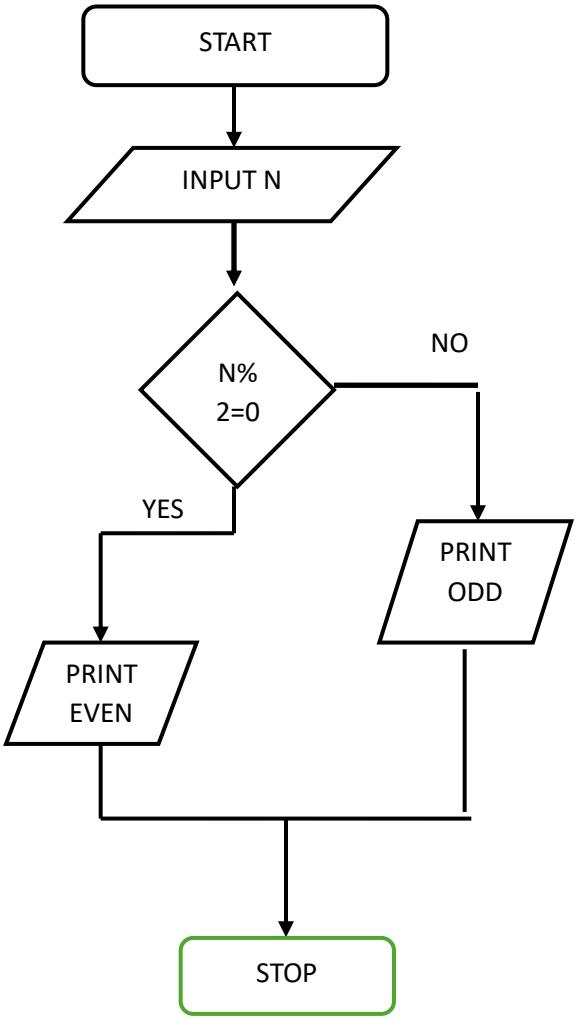
Step 3: Calculate Remainder = N % 2

Step 4: If Remainder is equal to 0, print "number is even"

Step 5: Else Print "Number is odd"

Step 6: Stop.

Flowchart :



Code:

```
#include <stdio.h>
#include <conio.h>

void main() {
   int N,remainder;

 clrscr();

 printf("Enter an integer: ");
 scanf("%d", &N);

remainder = N % 2;

if (remainder == 0) {
 printf("Number is even\n");
 } else {
   printf("Number is odd\n");
}

getch();
}
```

Output:

```
enter an integer:12
12 is an EVEN number.
```



10. Find out sum of digits of a given number.

Algorithm:

Step 1: Start

Step 2: Declare variable N

Step 3: Initialize sum = 0

Step 4: Repeat steps 5 to 7 while N > 0

Step 5: Extract the last digit: $\text{Digit} = N \% 10$

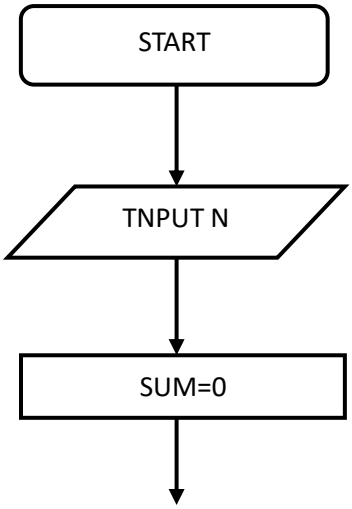
Step 6: Add digit to sum: $\text{sum} = \text{sum} + \text{Digit}$

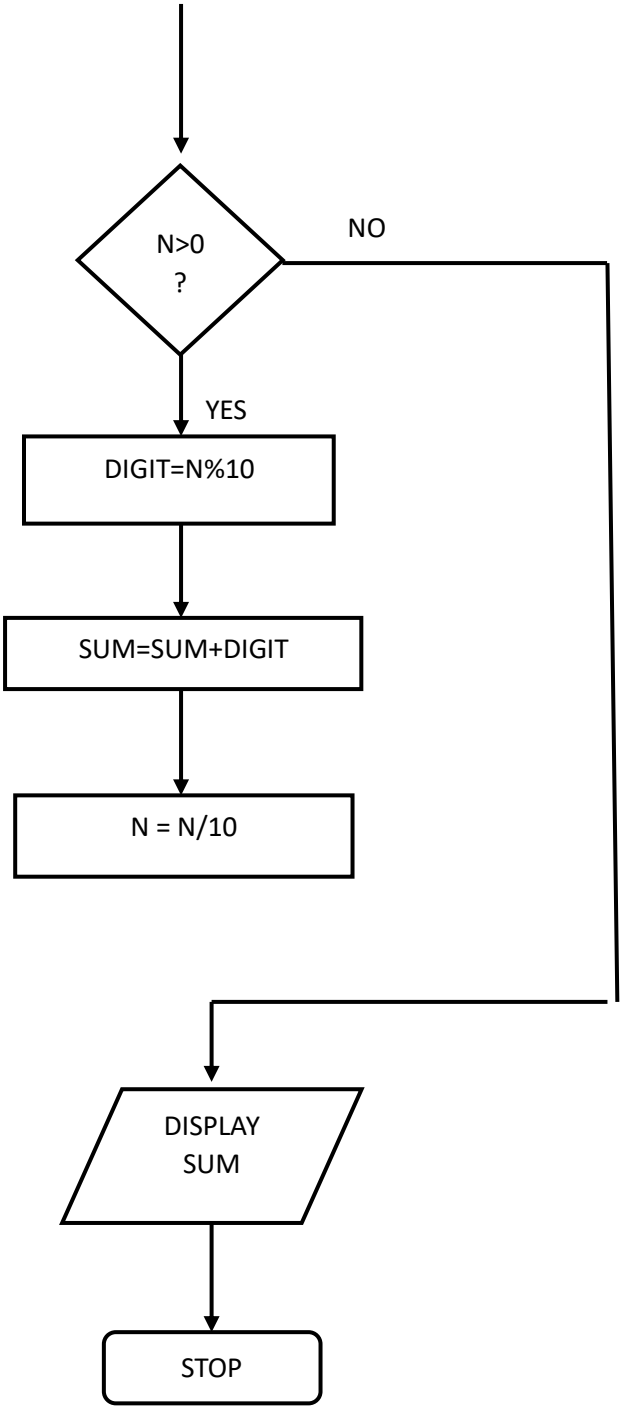
Step 7: Remove the last digit: $N = \text{Integer Div}$

Step 8: Print sum

Step 9: Stop

Flowchart:




N>0
?
NO
YES
DIGIT=N%10
SUM=SUM+DIGIT
N = N/10
DISPLAY
SUM
STOP

Code:

```
#include <stdio.h>
#include <conio.h>

void main() {
   int N, remainder, sum = 0;

   clrscr();

   printf("Enter an integer: ");
   scanf("%d", &N);

   if (num < 0) {
      N = -N;
   }

   while (num > 0) {
      remainder = N% 10;
      sum = sum+ remainder;
      N= N/ 10;
   }

   printf("Sum of digits = %d\n", sum);

   getch();
}
```

Output:



UNIT-2

1. PRINT”Hello world”

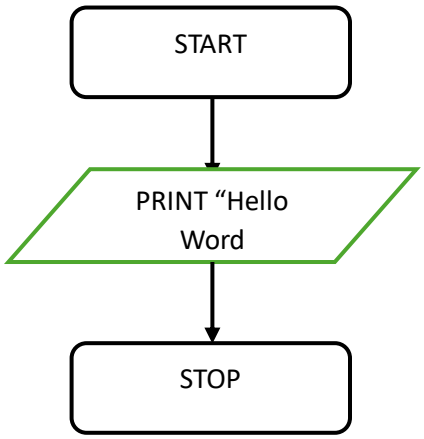
Algorithm:

Step 1 : START

Step 2 : PRINT “Hello World”

Step 3 : STOP

Flowchart:



Code:

```
#include<stdio.h>
#include<conio.h>

void main(){
        clrscr();
        printf("Hello World");
        getch();
}
```

Output:



2. Write a Program that takes 2 Values from user and Perform All Arithmetic Operation.

Algoritm:

Step 1 : START

Step 2 : DECLARE variables x=0.0, y=0.0, a, s, m, d

Step 3 : INPUT x,y

Step 4 : a = x + y

Step 5 : s = x – y

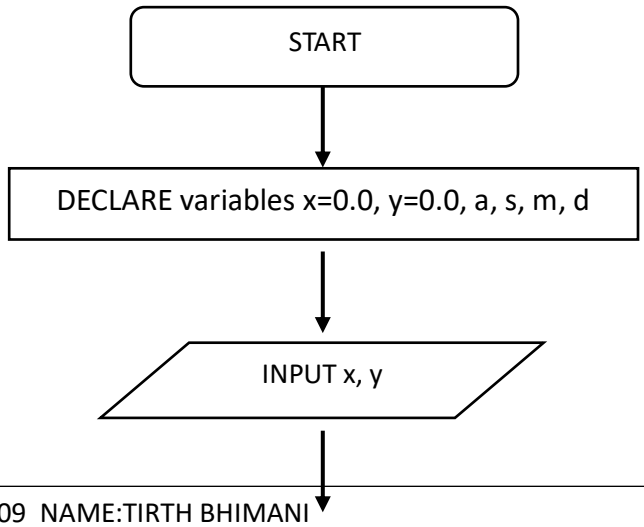
Step 6 : m = x * y

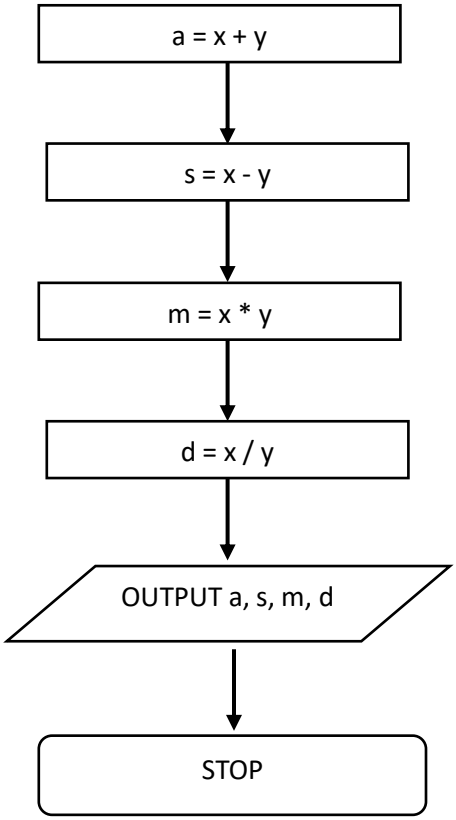
Step 7 : d = x / y

Step 8 : OUTPUT a, s, m, d

Step 9 : STOP

Flowchart:




a = x + y
s = x - y
m = x * y
d = x / y
OUTPUT a, s, m, d
STOP

Code:

```
#include<stdio.h>

#include<conio.h>


void main()

{

        float x,y;

        clrscr();

        printf("\n Enter Number 1: ");

        scanf("%f",&x);

        printf("\n Enter Number 2: ");

        scanf("%f",&y);


        printf(“Addition Is: %.2f \n”,x+y);

        printf(“Subtraction Is: %.2f \n”,x-y);

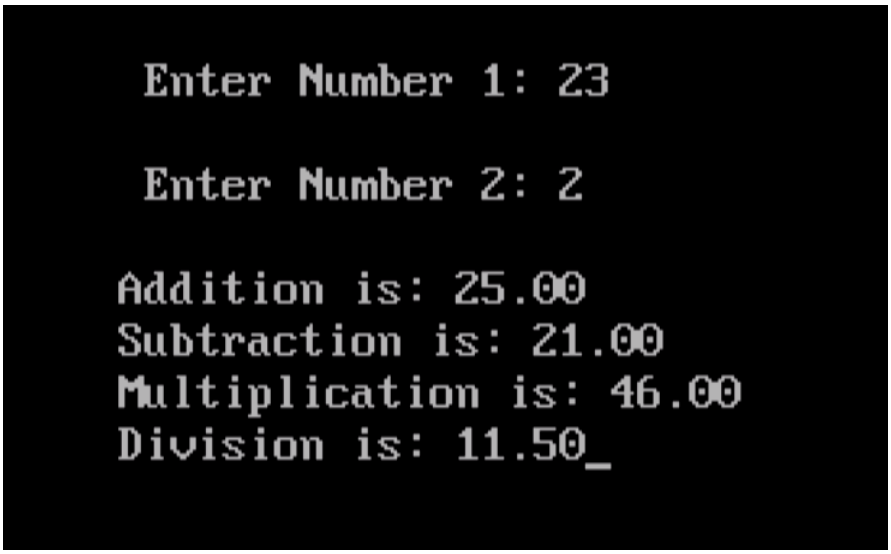
        printf(“Multiplication Is: %.2f \n”,x*y);

        printf(“Division Is: %.2f \n”,x/y);

        getch();

}
```

Output:


Enter Number 1: 23
Enter Number 2: 2
Addition is: 25.00
Subtraction is: 21.00
Multiplication is: 46.00
Division is: 11.50_

3. Write a program that takes 3 values for Principle Amount, Rate of Interest and No. of Years and find out Simple Interest.

Algorithm:

Step 1 : START

Step 2 : DECLARE variables SI, P, R, T

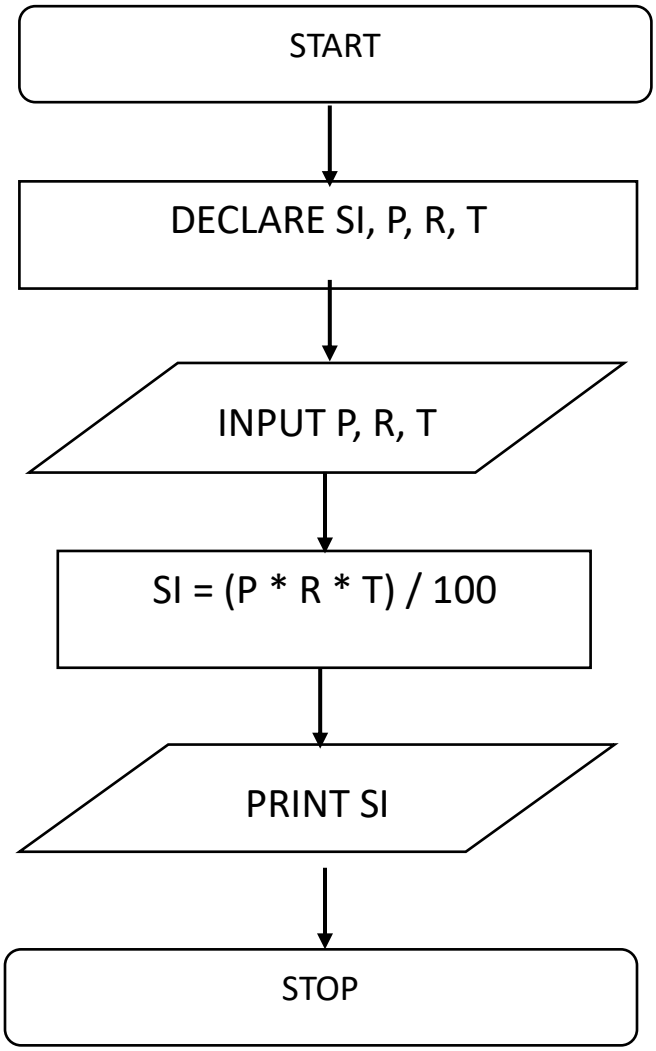
Step 3 : INPUT P, R, T

Step 4 : SI = (P * R * T) / 100

Step 5 : PRINT SI

Step 6 : STOP

Flowchart:



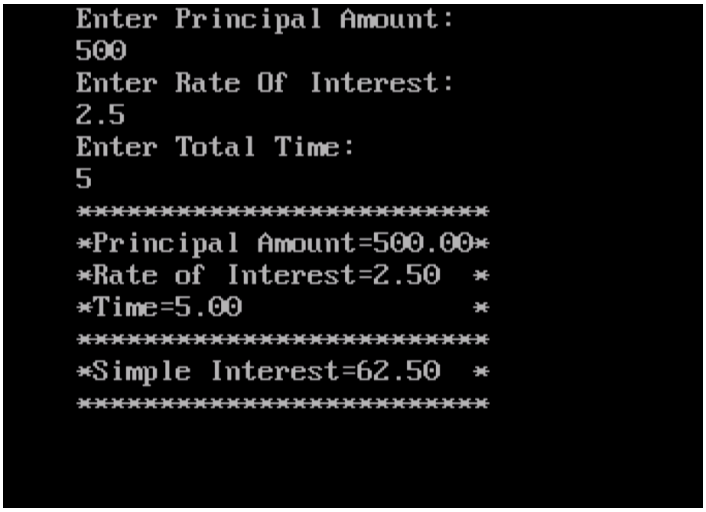
Code:

```
#include<stdio.h>
#include<conio.h>

void main(){
        float SI,P,R,T;
        clrscr();
        printf("Enter Principal Amount: \n");
        scanf("%f",&P);
        printf("Enter Rate Of Interest: \n");
        scanf("%f",&R);
        printf("Enter Total Time: \n");
        scanf("%f",&T);
        SI=(P*R*T)/100;
        printf("**************************\n");
        printf("*Principal Amount=%.2f*\n",P);
        printf("*Rate of Interest=%.2f  *\n",R);
        printf("*Time=%.2f            *\n",T);
        printf("**************************\n");
        printf("*Simple Interest=%.2f  *\n",SI);
        printf("**************************");

        getch();
}
```

Output:



4. Write a program that input value from user and find out Area of Circle.

Algorithm:

Step 1 : START

Step 2 : DECLARE variable r, area

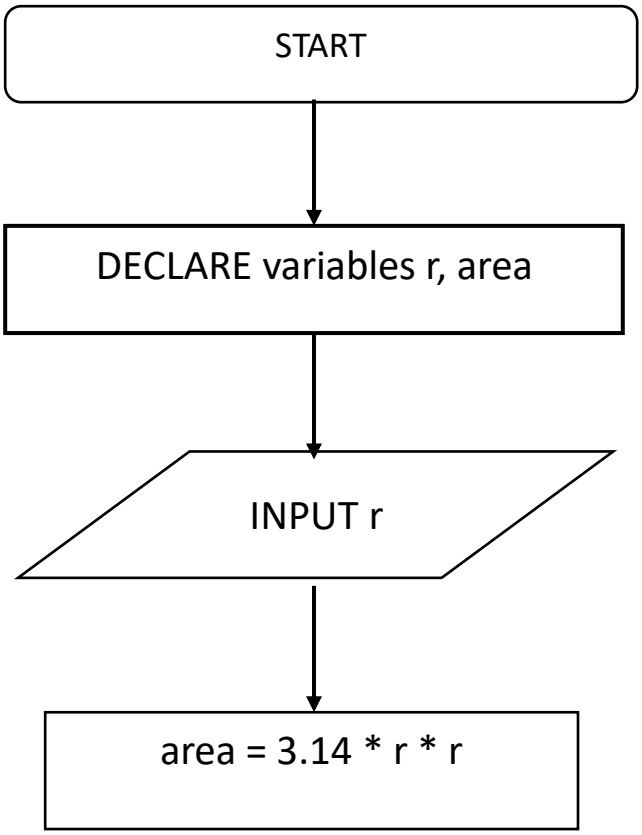
Step 3 : INPUT r

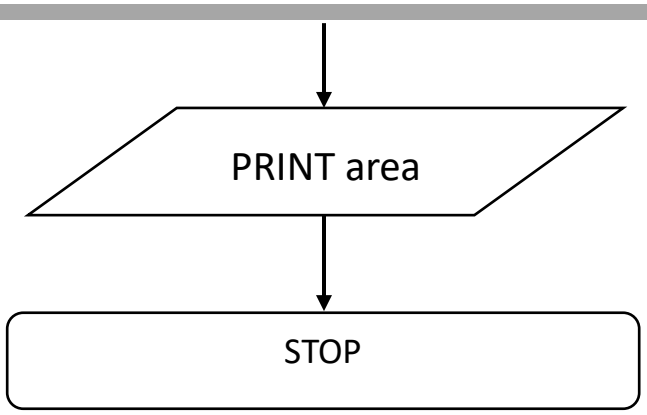
Step 4 : area = 3.14 * r * r

Step 5 : PRINT area

Step 6 : STOP

Flowchart:



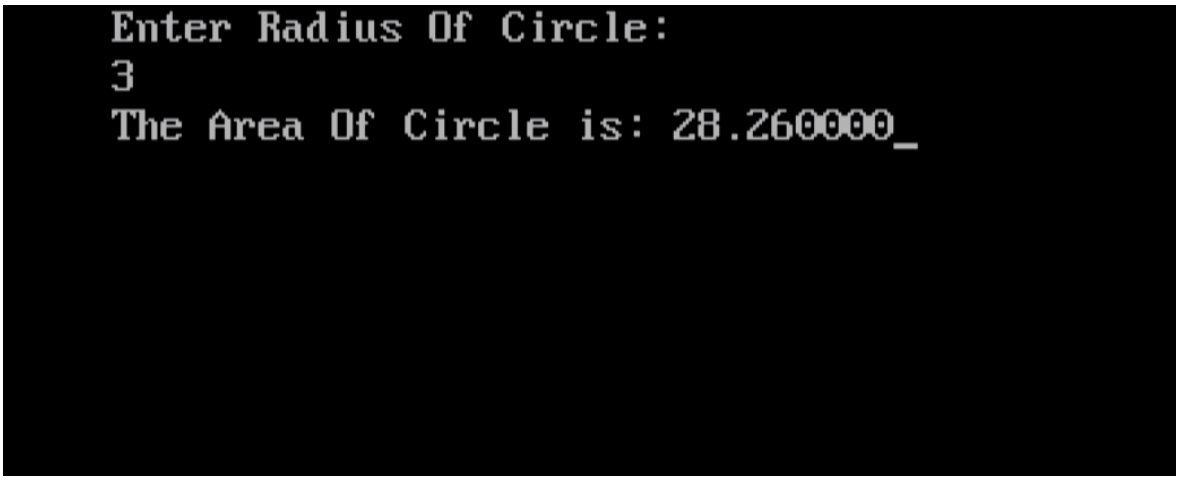

PRINT area
STOP

Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
        float Radius;
        clrscr();
        printf("Enter Radius Of Circle: \n");
        scanf("%f",&Radius);
        printf("The Area Of Circle is: %f",3.14*Radius*Radius);
        getch();
}
```

Output:



5. Write a program that input two values from user and find out Area of Rectangle.

Algorithm:

Step 1 : START

Step 2 : DECLARE variable l, b, area

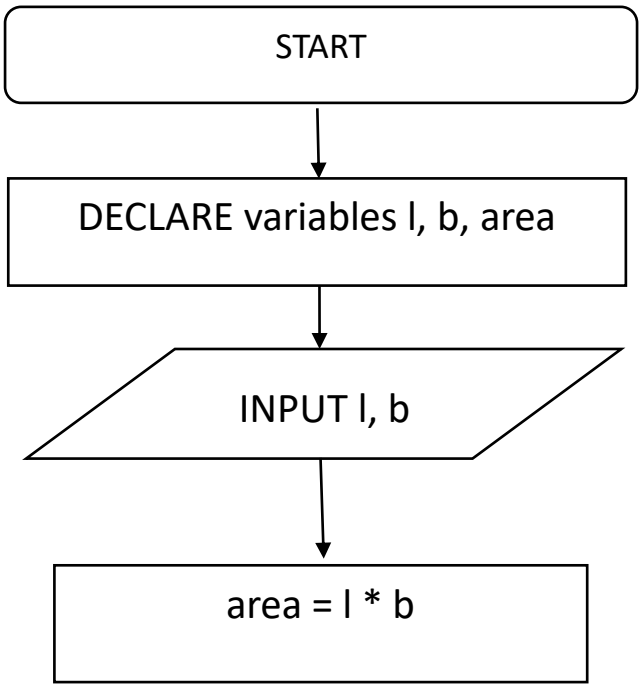
Step 3 : INPUT l, b

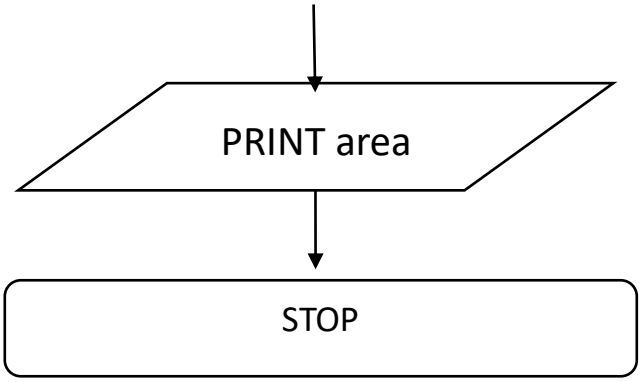
Step 4 : area = l * b

Step 5 : PRINT area

Step 6 : STOP

Flowchart:



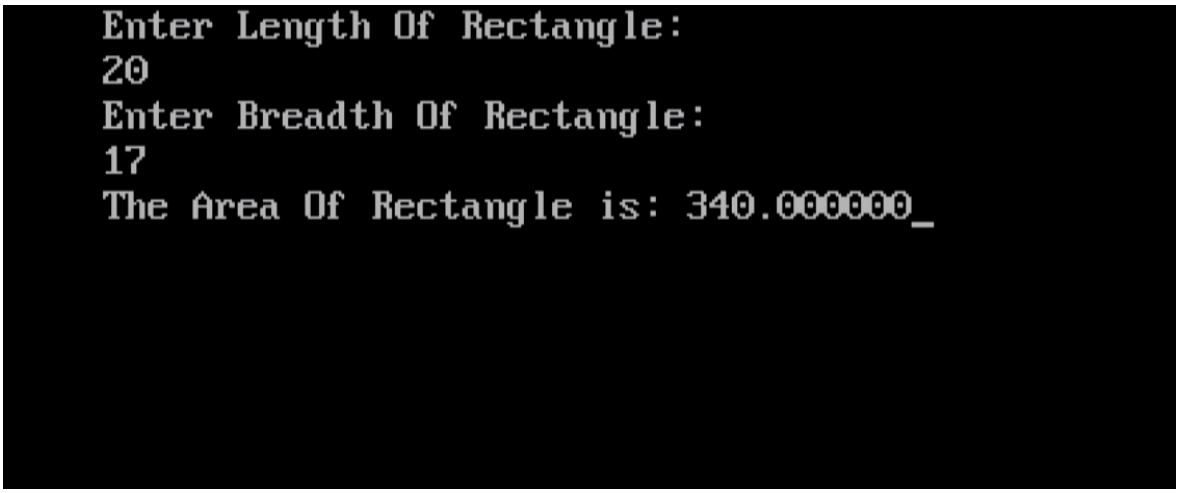


Code:

```
#include<stdio.h>
#include<conio.h>

void main()
{
        float Length,Breadth;
        clrscr();
        printf("Enter Length Of Rectangle: \n");
        scanf("%f",&Length);
        printf("Enter Breadth Of Rectangle: \n");
        scanf("%f",&Breadth);
        printf("The Area Of Rectangle is: %f",Length*Breadth);
        getch();
}
```

Output :



6. Write a program that input 3 values from user and find out its Average

---

Algorithm:

Step 1 : START

Step 2 : DECLARE variables x, y, z, avg

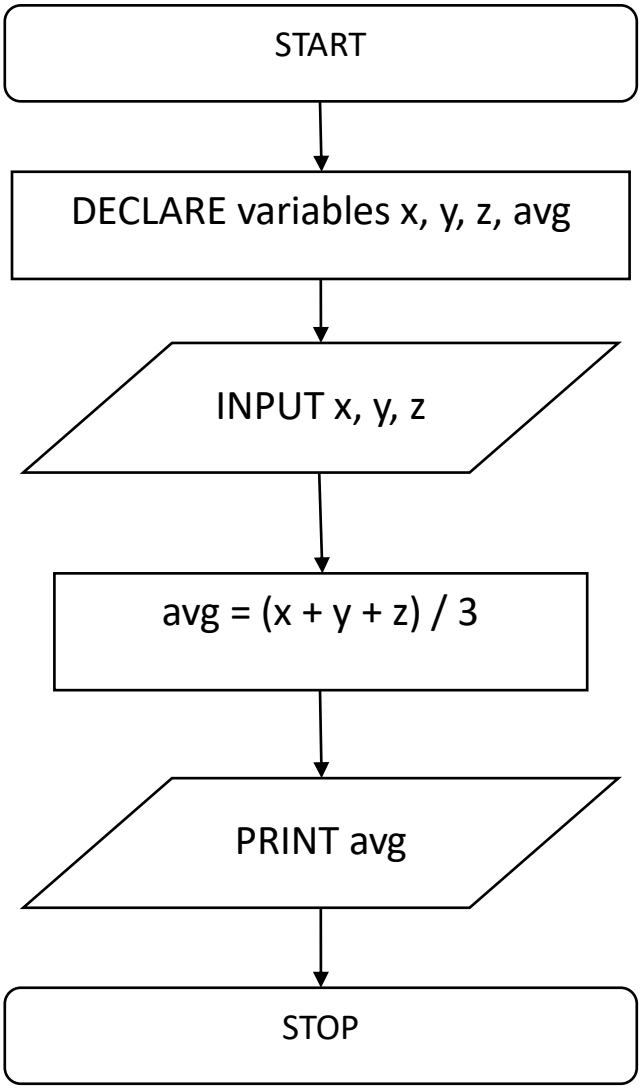
Step 3 : INPUT x, y, z

Step 4 : avg = (x + y + z) / 3

Step 5 : PRINT avg

Step 6 : STOP

Flowchart:



Code:

```
#include<stdio.h>

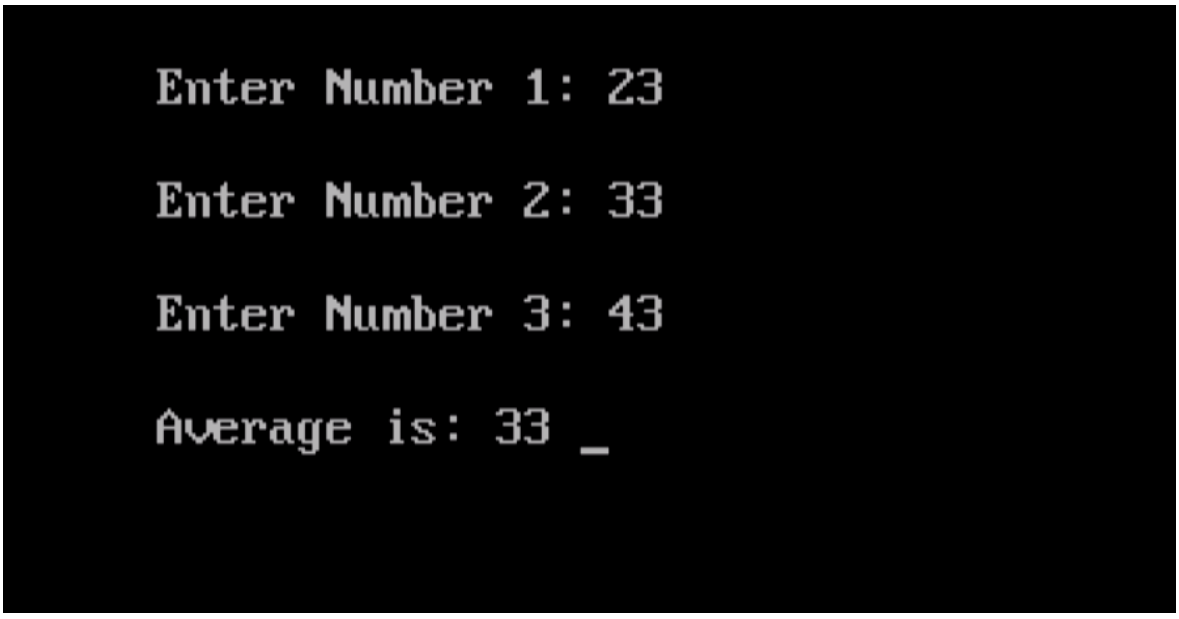
#include<conio.h>


void main()
```

```
{
                        int x,y,z;
                clrscr();
        printf("\n Enter Number 1: ");
        scanf("%d",&x);
        printf("\n Enter Number 2: ");
        scanf("%d",&y);
         printf("\n Enter Number 3: ");
        scanf("%d",&z);
        printf("\n Average is: %d ",(x+y+z)/3);
        getch();

}
```

Output:



7. Write a program to Calculate the Square and Cube of a number.

Algorithm:

Step 1 : START

Step 2 : DECLARE variables n, sqr, cube

Step 3 : INPUT n

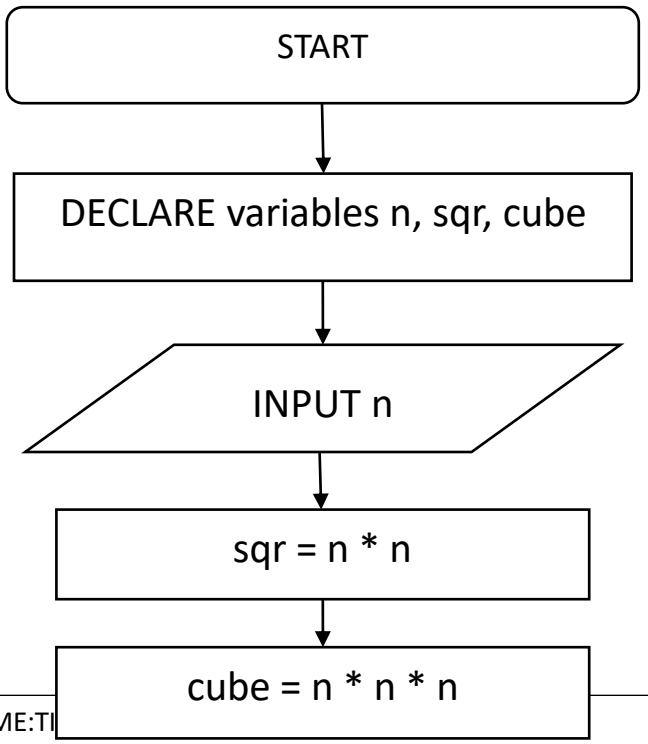
Step 4 : sqr = n * n

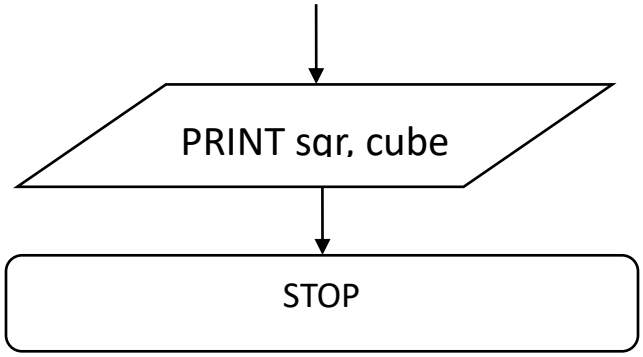
Step 5 : cube = n* n* n

Step 6 : OUTPUT sqr, cube

Step 7 : STOP

Flowchart:





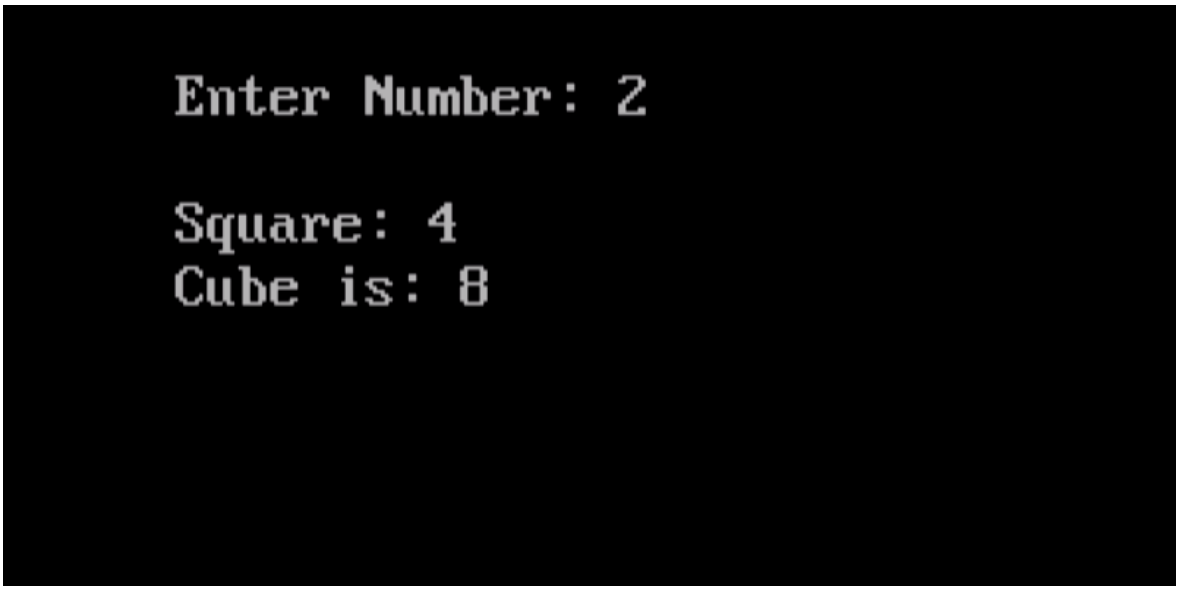
Code:

```
#include<stdio.h>
#include<conio.h>

void main()
{
        int n;
        clrscr();
        printf("\n Enter Number: ");
        scanf("%d",&n);
        printf("\n Square is: %d",n*n);
        printf("\n Cube is: %d",n*n*n);
        getch();

}
```

Output:



8. Write a program Interchange the value of two numbers using third variable

---

Flowchart:

Step 1 : START

Step 2 : DECLARE variables temp, x, y

Step 3 : INPUT x, y

Step 4 : temp = x

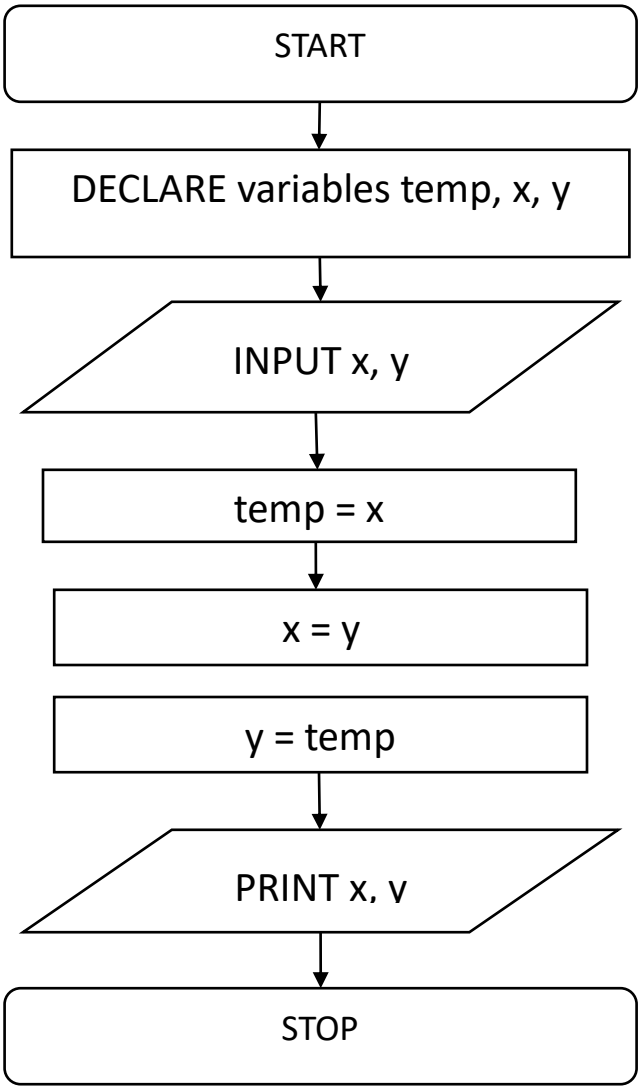
Step 5 : x = y

Step 6 : y = temp

Step 7 : PRINT x, y

Step 8 : STOP

Flowchart:



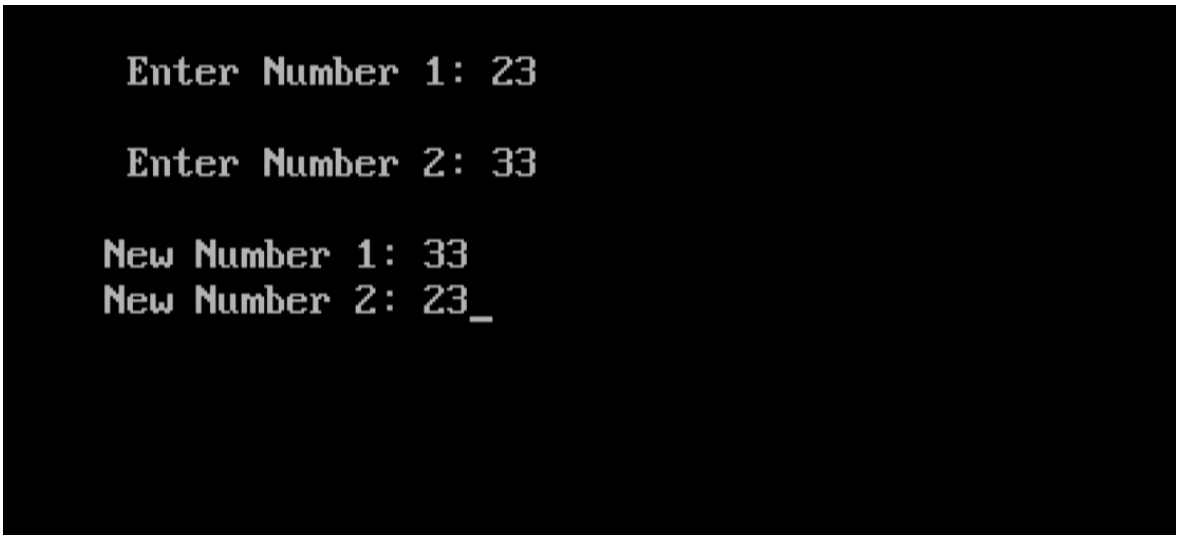
Code:

```
#include<stdio.h>
#include<conio.h>

void main()
{
        int temp,x,y;
        clrscr();
        printf("\n Enter Number 1: ");
        scanf("%d",&x);
        printf("\n Enter Number 2: ");
        scanf("%d",&y);
        temp=x;
        x=y;
        y=temp;
        printf("\nNew Number 1: %d",x);
        printf("\nNew Number 2: %d",y);
        getch();

}
```

Output:



9. Write a program Interchange the value of two numbers without using third variable

Algorithm:

Step 1 : START

Step 2 : DECLARE variables q, pr, disc, f_amt

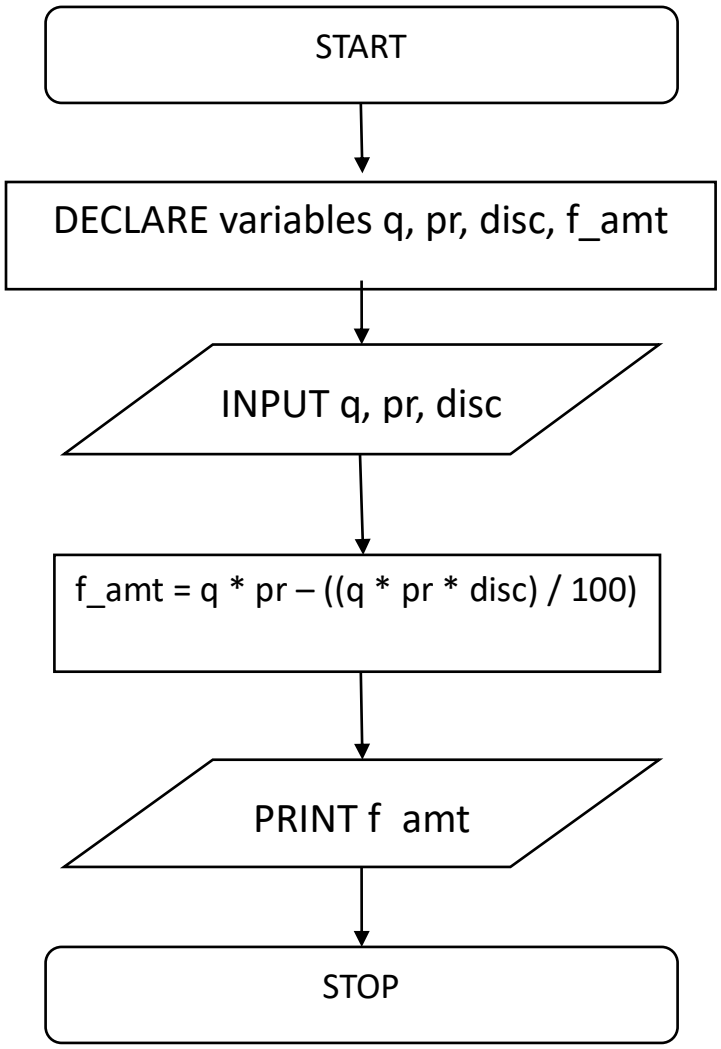
Step 3 : INPUT q, pr, disc

Step 4 : f_amt = q * pr – ((q * pr * disc) / 100)

Step 5 : OUTPUT f_amt

Step 6 : STOP

Flowchart:


START
DECLARE variables q, pr, disc, f_amt
INPUT q, pr, disc
f_amt = q * pr – ((q * pr * disc) / 100)
PRINT f amt
STOP

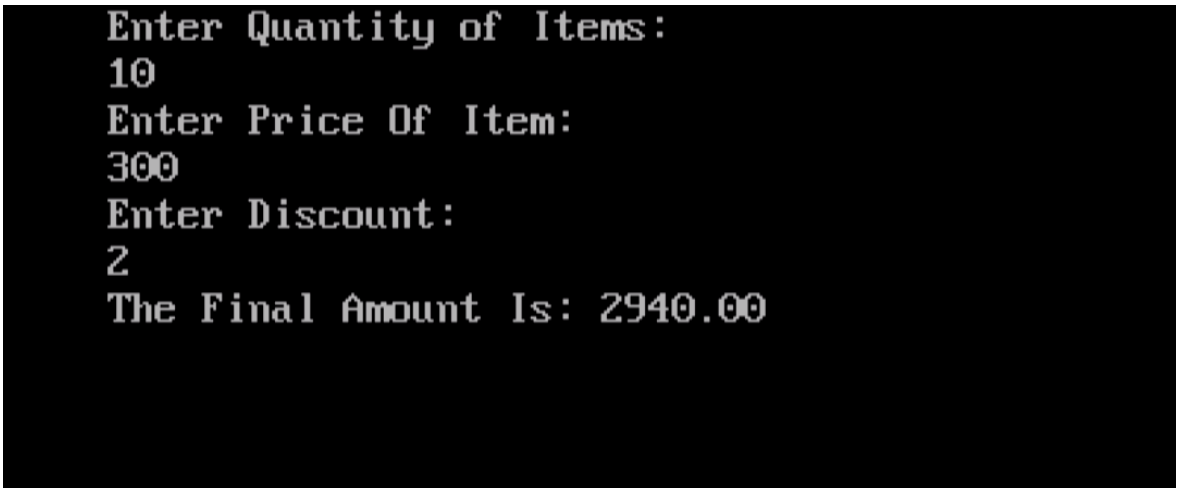
Code:

```
#include<stdio.h>
#include<conio.h>

void main(){

    float Input_quantity,Price,Discount,Final_Amount;
    clrscr();
    printf("Enter Quantity of Items: \n");
    scanf("%f",&Input_quantity);
    printf("Enter Price Of Item: \n");
    scanf("%f",&Price);
    printf("Enter Discount: \n");
    scanf("%f",&Discount);
    Final_Amount = Input_quantity*Price-((Input_quantity*Price*Discount)/100);
    printf("The Final Amount Is: %.2f",Final_Amount);
    getch();
}
```

Output :


Enter Quantity of Items:
10
Enter Price Of Item:
300
Enter Discount:
2
The Final Amount Is: 2940.00

	Unit-3
1.	Write a program that input 2 values from user and find out Maximum.
2.	Write a program that input 2 values from user and find out Minimum.
3.	Write a program that input age from user if age>=18 then print person is eligible for vote else person is not eligible for vote.
4.	Write a program that input value from user and find out number is even or odd.
5.	Write a program that input 2 number from user and find out number is equal or not.
6.	Write a program that Input Salary From the user if salary Greater than or equal to 5000 then hr=5% of basic salary, ta=6% of basic salary, da=4% of basic salary and pf=5% of basic salary. but if salary is less than 5000 then hra=4%,ta=5%,da=3% and pf=4%.find out gross salary. printhra,ta,da,pf and gross salary(Hint : take all variable as float. Formula for gross salary=salary+hra+ta+da- pf).
7.	Write a C Program to input price quantity of an item from user and display final payment according to following condition. If total purchase(price*quantity) is greater than Rs.1000 then calculate discount 15% otherwise calculate discount 10% of total rupees.
8.	Write a program to input year and find whether year is leap year or not.(Hint Use % Sign)
9.	Write a program to input two values from user and check whether the first number is divisible by second or not. (Hint Use % Sign)
10.	Write a program to Input a character, if it is capital convert into small else convert into capital.
11.	Write a program that input one character from user and find out that character is vowel or not.(aeiou is called vowel).
12.	Write programs that find out character is in uppercase or lowercase. Using If...Else..If or Switch.. Case
13.	Input Three Values from user and Find Maximum.
14.	Input Three Values from user and Find Minimum.

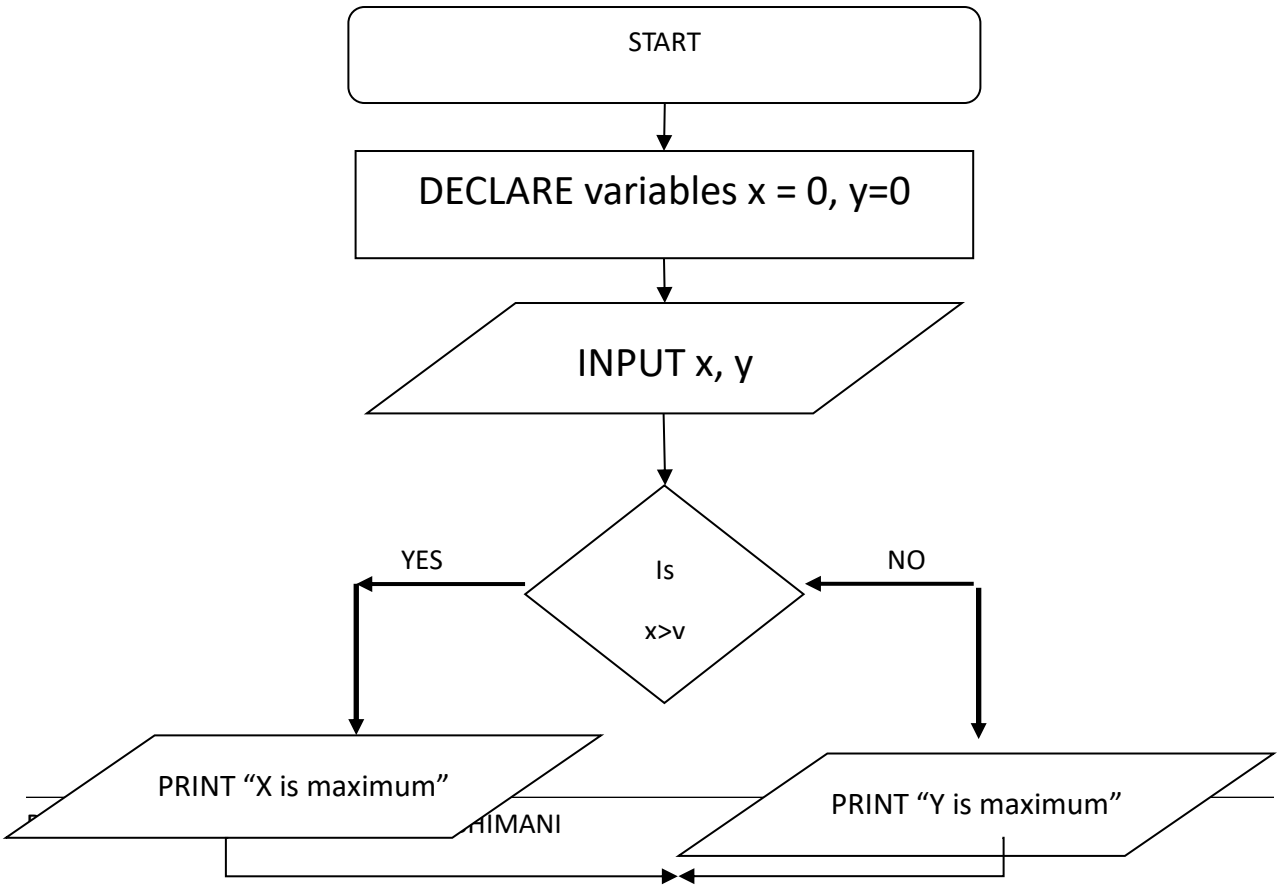
15.	Input RollNumber, Marks of 5 Subjects and find out Total, Percentage, Result and Grade.
16.	Input day in number and print in text(for Example input 1 then Output Monday)
17.	Input Month in number and print in text(for example input 1 then output January)
18.	Input number and check whether the number is positive, negative or zero.

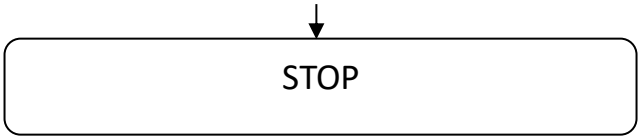
1. Write a program that input 2 values from user and find out Maximum.

Algorithm:

Step 1 : START
Step 2 : DECLARE variables num1, num2, maximum
Step 3 : INPUT num1, num2
Step 4 : IF num1 > num2 THEN maximum = num1
Step 5 : ELSE IF num2 > num1 THEN maximum = num2
Step 6 : ELSE DISPLAY "Both numbers are equal" AND GO TO STEP 9
Step 7 : OUTPUT maximum
Step 8 : STOP

Flowchart:




STOP

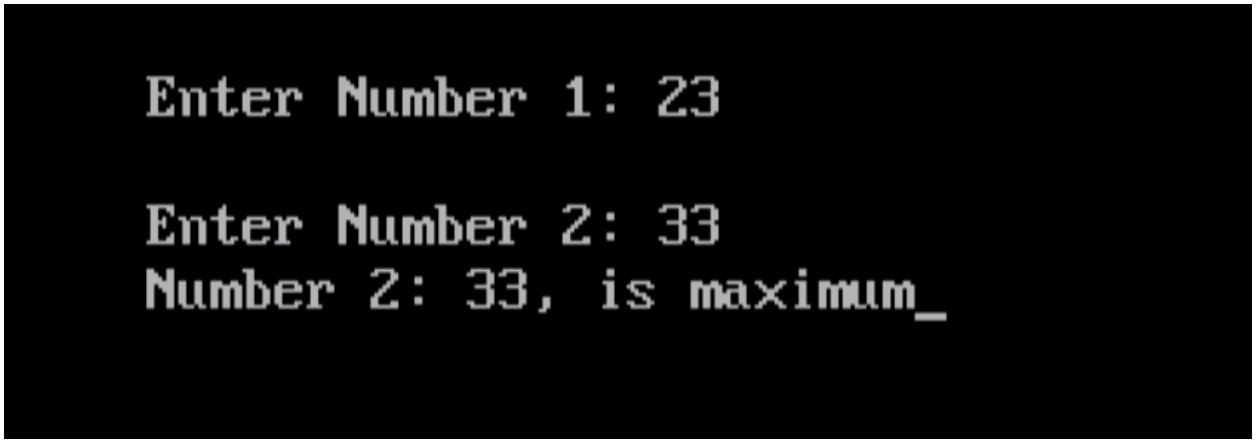
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int x,y;
   clrscr();
   printf("\nEnter Number 1: ");
   scanf("%d",&x);
   printf("\nEnter Number 2: ");
   scanf("%d",&y);
   if(x!=y){
      if(x>y){
                        printf("Number 1: %d, is maximum", x);
      }
      else
                        printf("Number 2: %d, is maximum", y);
   }
   else{
                     printf("Both are equal");
   }
```

```
    getch();
}
```

Output :



(2) Write a program that input 2 values from user and find out Minimum.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x=0, y=0

Step 3 : INPUT x, y

Step 4 : IF x < y THEN

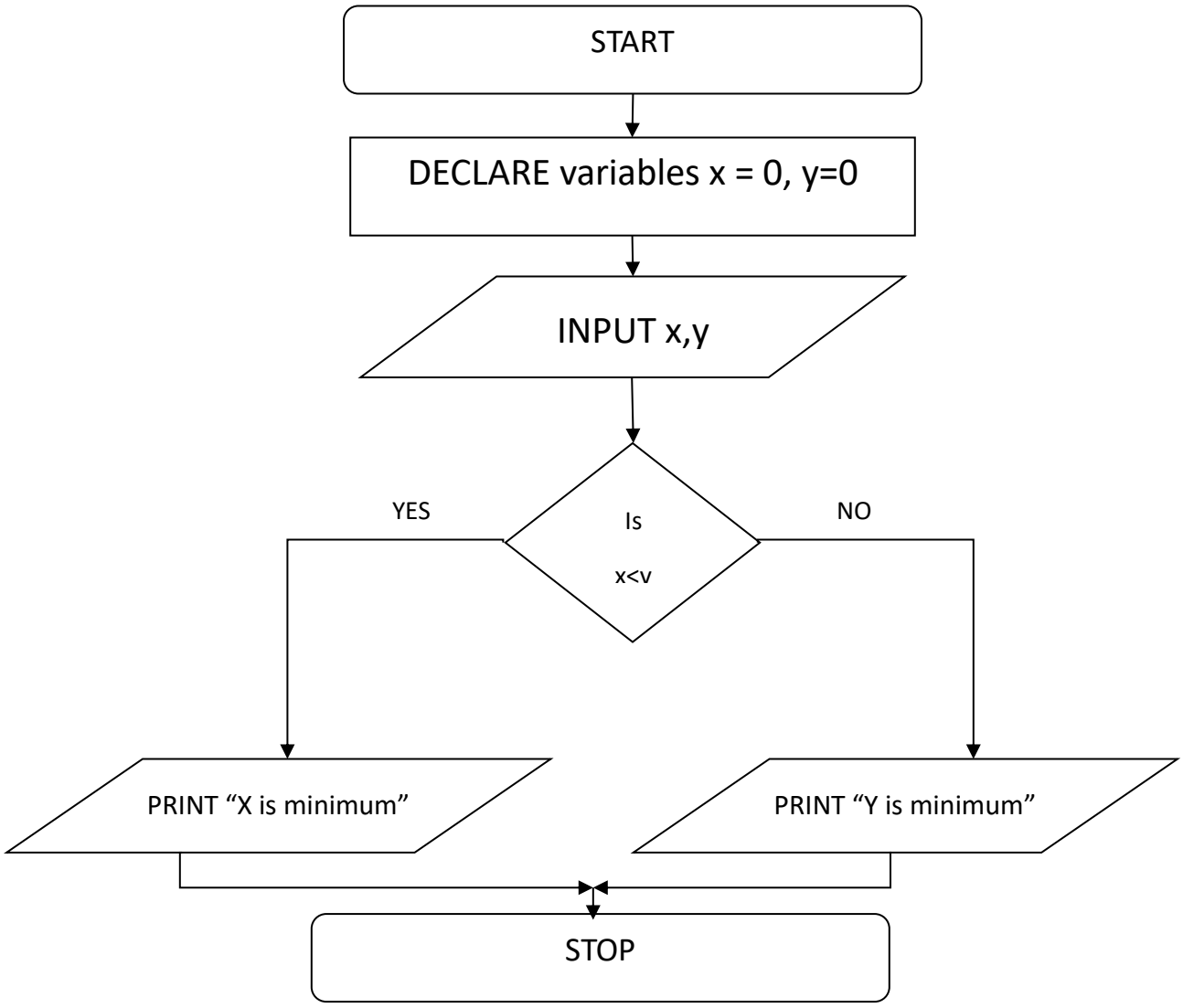
PRINT "X is Minimum"

Step 5 : ELSE

PRINT "Y is Minimum"

Step 6 : STOP

Flowchart :

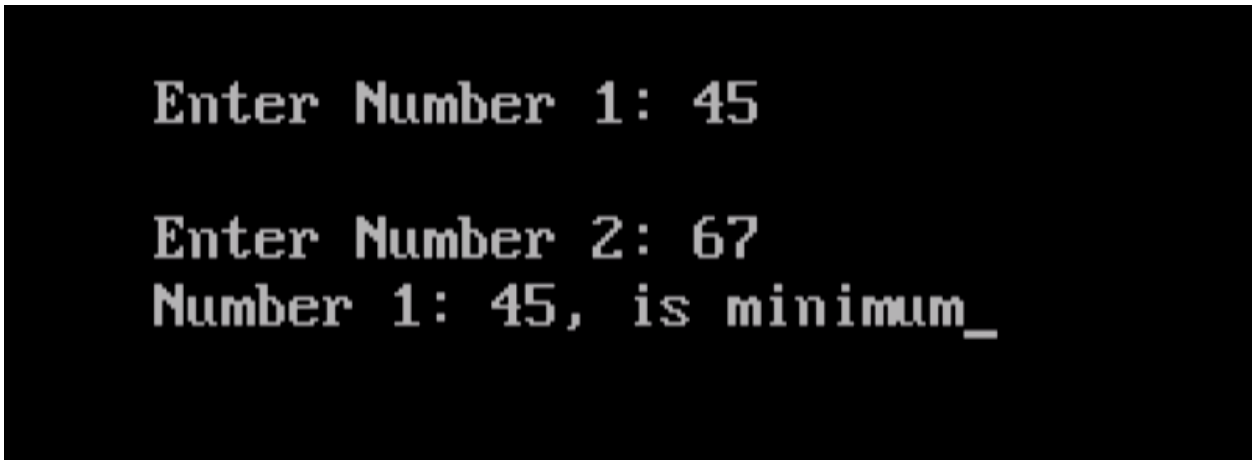

START
DECLARE variables x = 0, y=0
INPUT x,y
Is
x<v
YES
NO
PRINT “X is minimum”
PRINT “Y is minimum”
STOP

Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int x,y;
   clrscr();
   printf("\nEnter Number 1: ");
   scanf("%d",&x);
   printf("\nEnter Number 2: ");
   scanf("%d",&y);
   if(x<y){
      printf("Number 1: %d, is minimum", x);
   }
   else if(y<x)
                  printf("Number 2: %d, is minimum", y);
   else
                  printf("Numbers are equal");
   getch();
}
```

Output :



(3) Write a program that input age from user if age>=18 then print person is eligible for vote else person is not eligible for vote.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable age=0

Step 3 : INPUT age

Step 4 : IF age >= 18 THEN

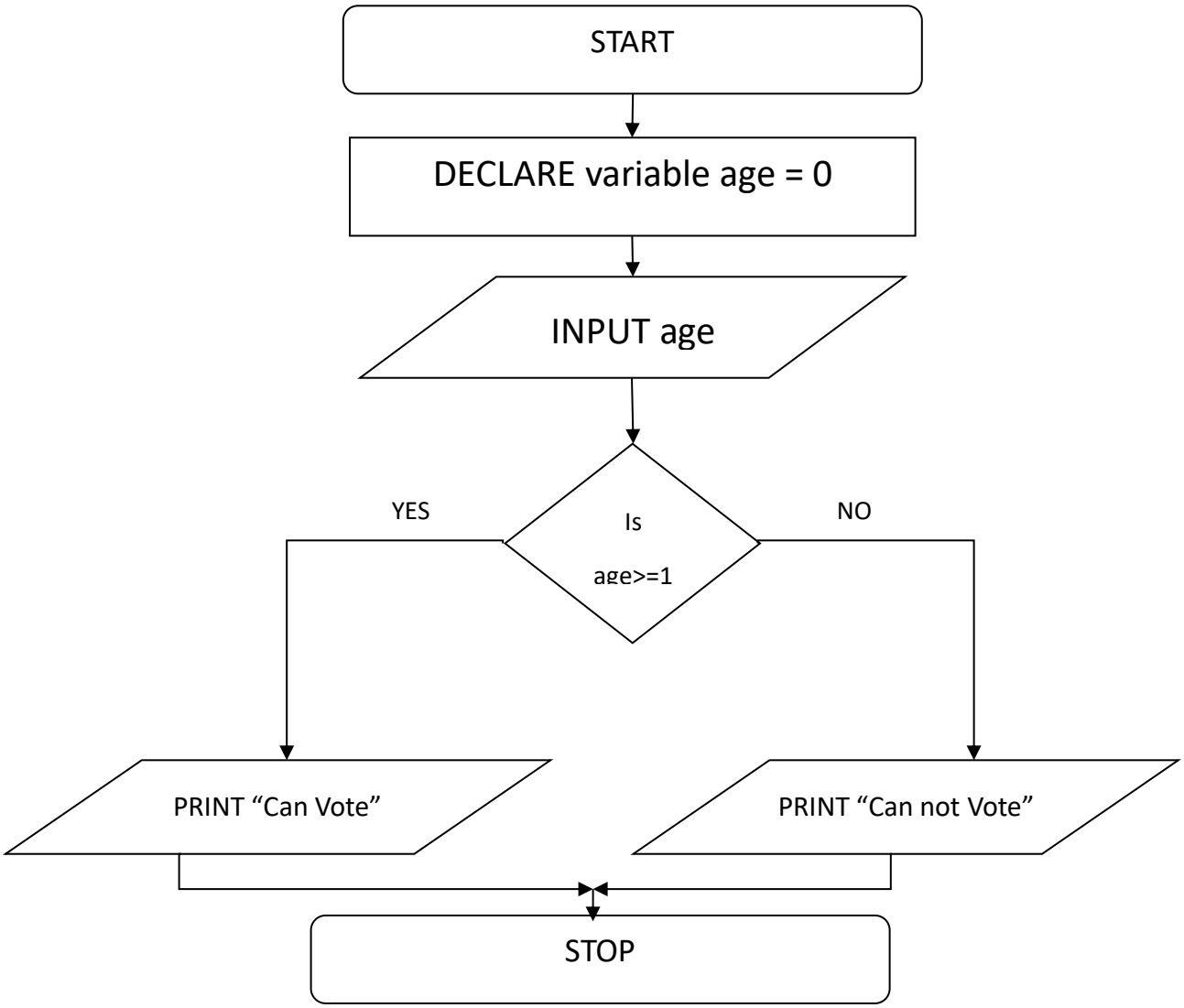
PRINT "Person is eligible to vote"

Step 5 : ELSE

PRINT "Person is not eligible to vote"

Step 6 : STOP

Flowchart :

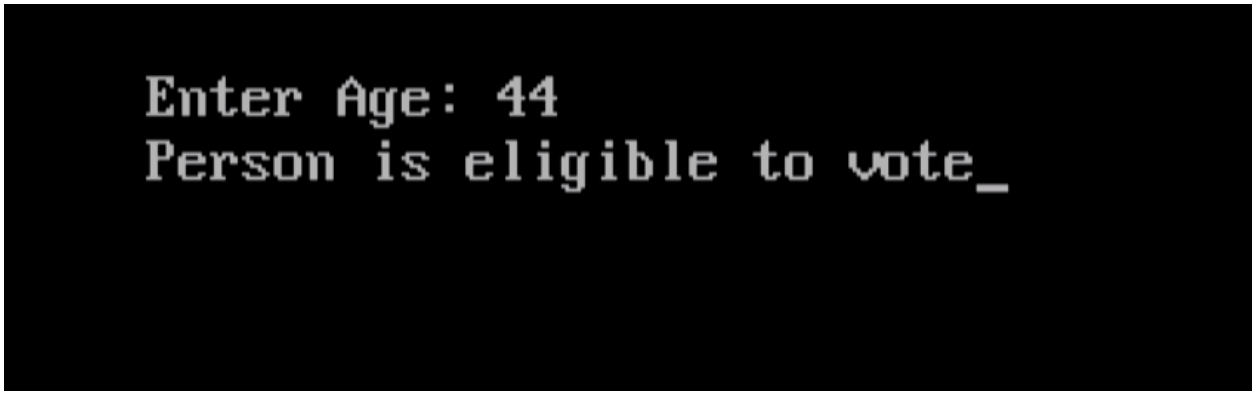

START
DECLARE variable age = 0
INPUT age
Is
age>=1
YES
NO
PRINT “Can Vote”
PRINT “Can not Vote”
STOP

Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int x;
   clrscr();
   printf("\nEnter Age: ");
   scanf("%d",&x);
   if(x>=18){
      printf("Person is eligible to vote");
   }
   else
      printf("Person is not eligible to vote");
   getch();
}
```

Output:



(4) Write a program that input value from user and find out number is even or odd.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable x=0

Step 3 : INPUT x

Step 4 : IF x % 2 == 0 THEN

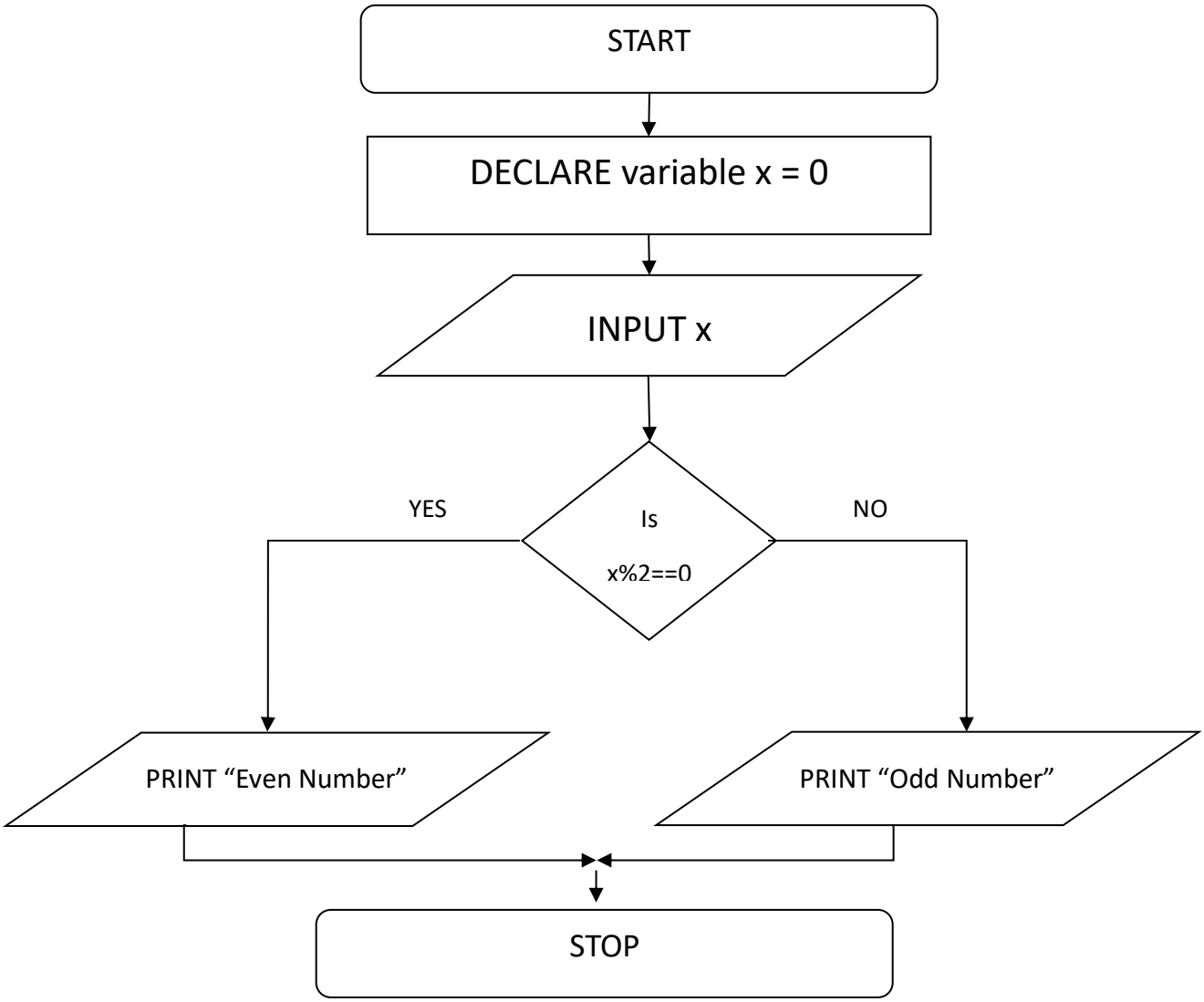
PRINT "Even Number"

Step 5 : ELSE

PRINT "Odd Number"

Step 6 : STOP

Flowchart :

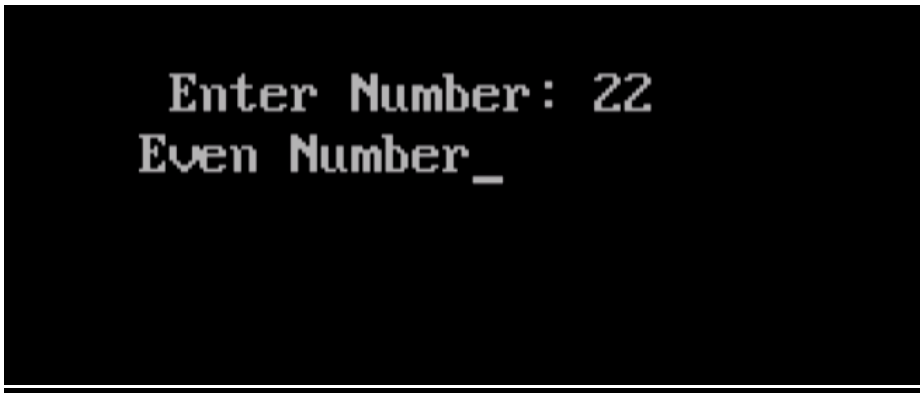

START
DECLARE variable x = 0
INPUT x
Is
x%2==0
YES
NO
PRINT “Even Number”
PRINT “Odd Number”
STOP

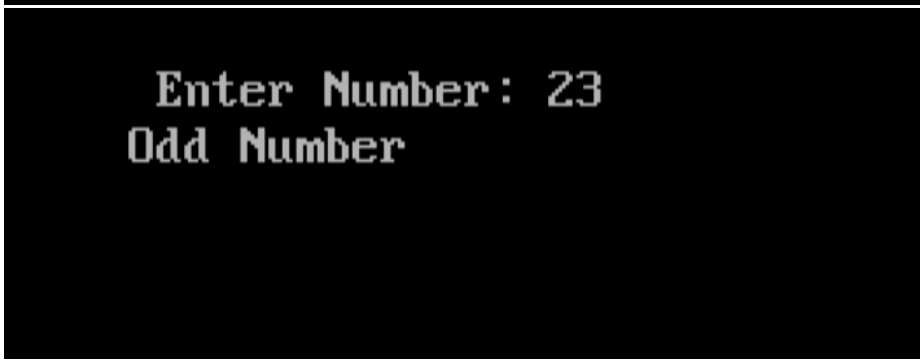
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int x;
   clrscr();
   printf("\nEnter Number: ");
   scanf("%d",&x);
   if(x%2==0){
      printf("Even Number");
   }
   else
      printf("Odd Number");
   getch();
}
```

Output :


Enter Number: 22
Even Number_


Enter Number: 23
Odd Number

(5) Write a program that input 2 number from user and find out number is equal or not.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x=0, y=0

Step 3 : INPUT x, y

Step 4 : IF x == y THEN

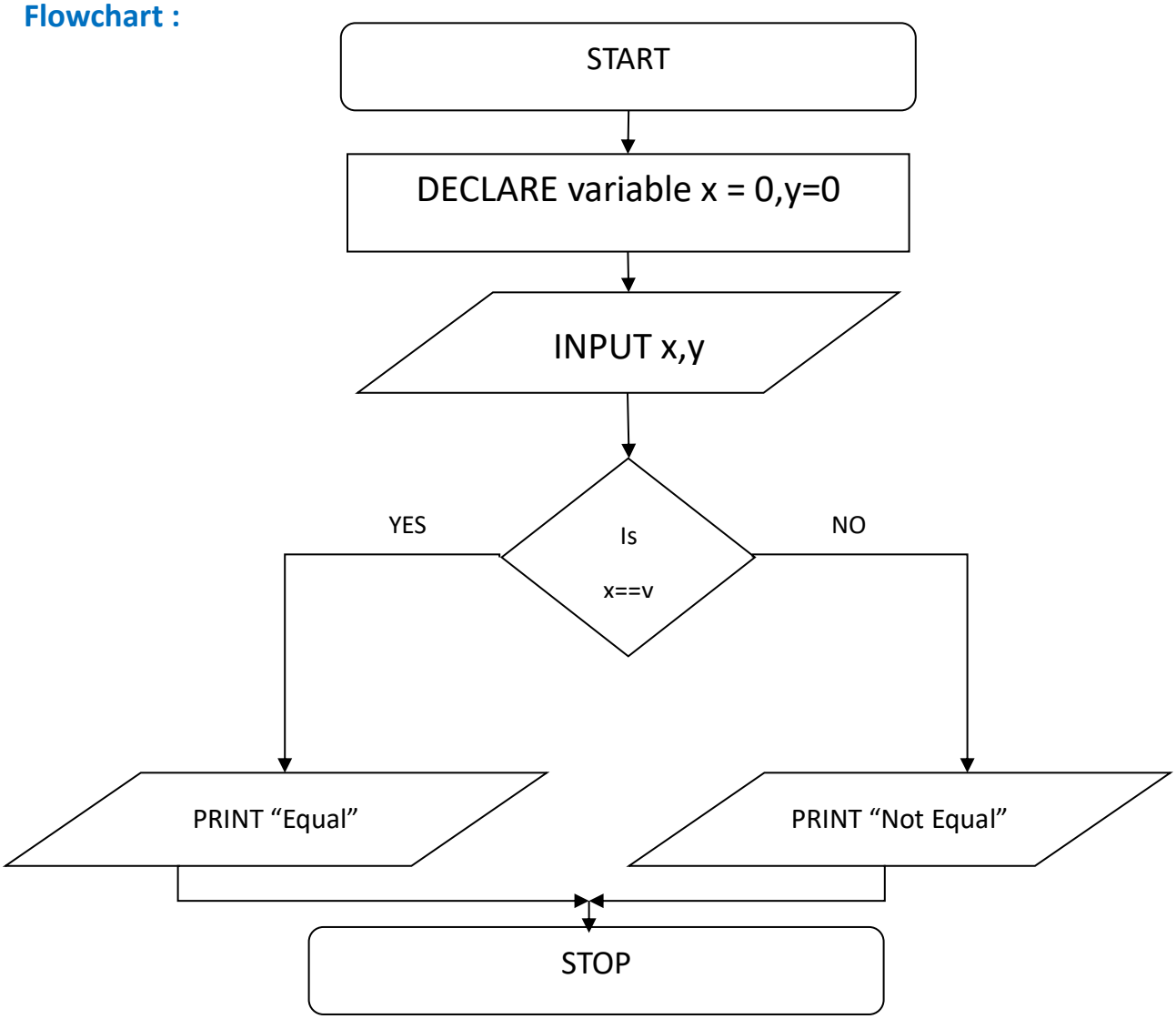
PRINT "Equal Numbers"

Step 5 : ELSE

PRINT "Not Equal Numbers"

Step 6 : STOP

Flowchart :

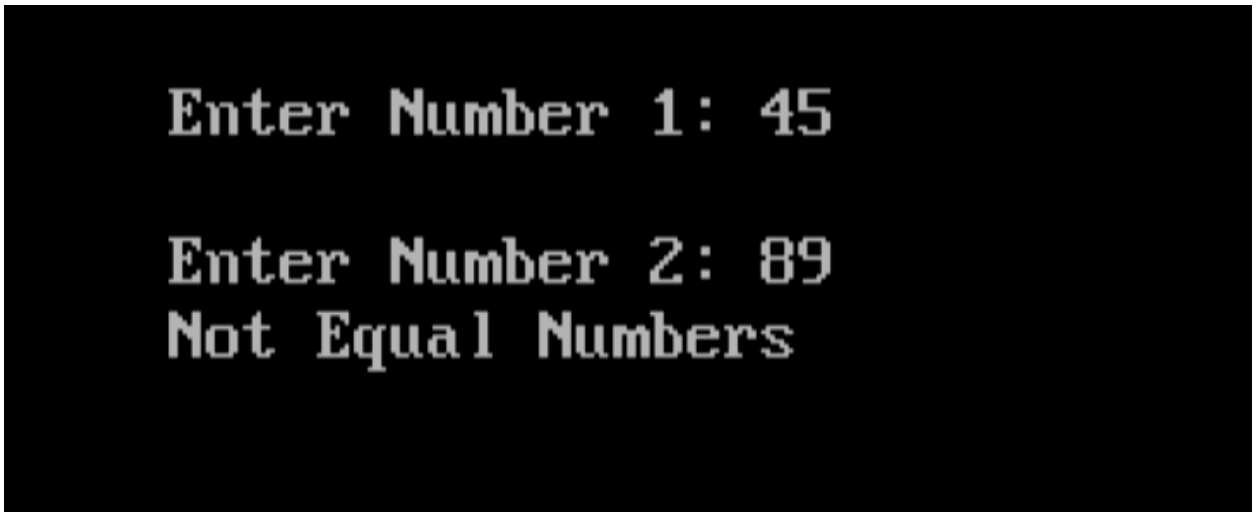

START
DECLARE variable x = 0,y=0
INPUT x,y
Is
x==v
YES
NO
PRINT “Equal”
PRINT “Not Equal”
STOP

Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int x,y;
   clrscr();
   printf("\nEnter Number 1: ");
   scanf("%d",&x);
   printf("\nEnter Number 2: ");
   scanf("%d",&y);
   if(x==y){
      printf("Equal Numbers");
   }
   else
      printf("Not Equal Numbers");
   getch();
}
```

Output :


Enter Number 1: 45
Enter Number 2: 89
Not Equal Numbers

(6) Write a program that Input Salary From the user if salary Greater than or equal to 5000 then hr=5% of basic salary, ta=6% of basic salary, da=4% of basic salary and pf=5% of basic salary. but if salary is less than 5000 then hra=4%,ta=5%,da=3% and pf=4%.find out gross salary. printhra,ta,da,pf and gross salary(Hint : take all variable as float. Formula for gross salary=salary+hra+ta+da- pf).

Algorithm :

Step 1 : START

Step 2 : DECLARE variables sal, hra, ta, da, pf, gross_sal

Step 3 : INPUT sal

Step 4 : IF sal >= 5000 THEN

hra = 5% of sal

ta = 6% of sal

da = 4% of sal

pf = 5% of sal

Step 5 : ELSE

hra = 4% of sal

ta = 5% of sal

da = 3% of sal

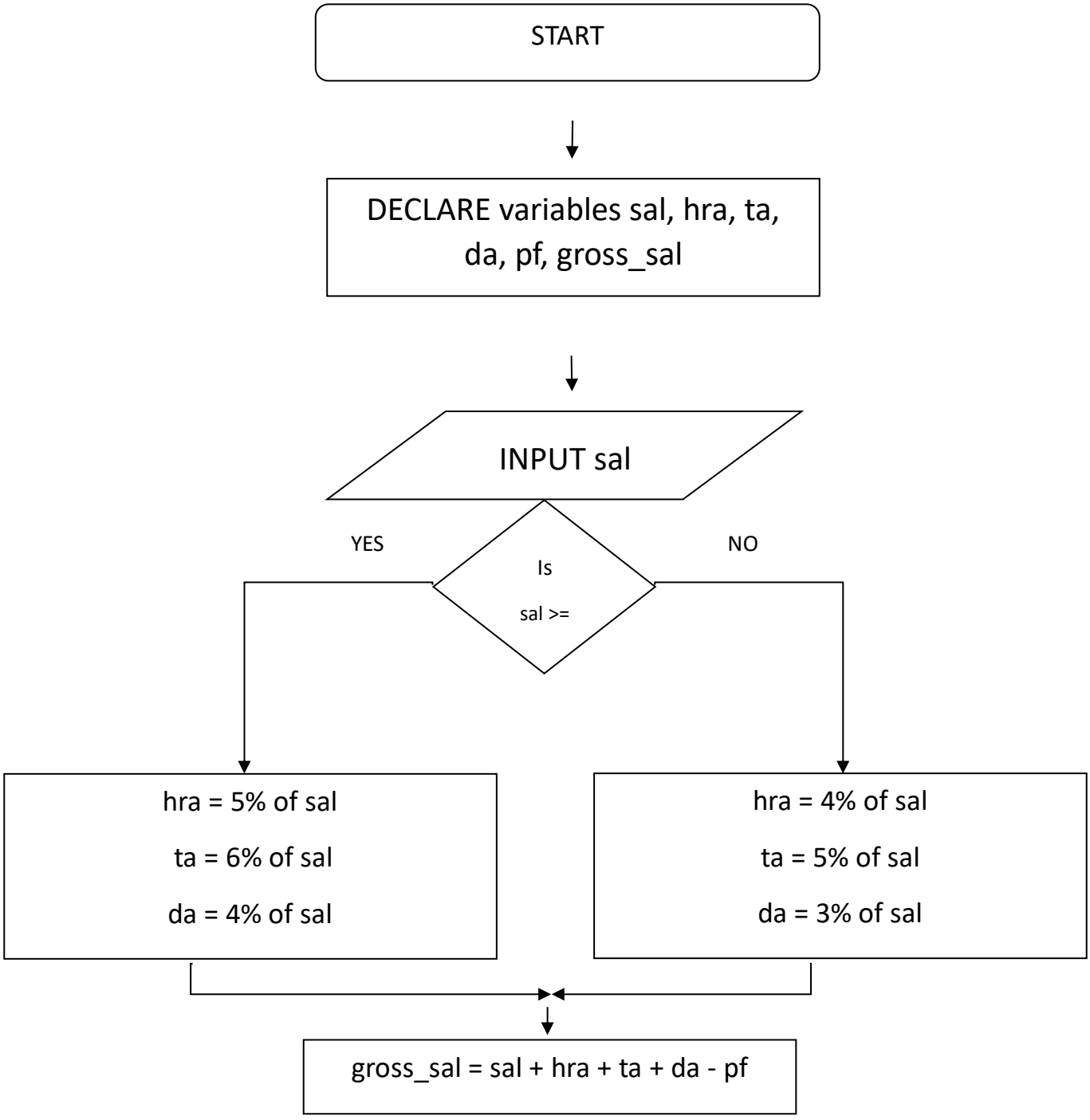
pf = 4% of sal

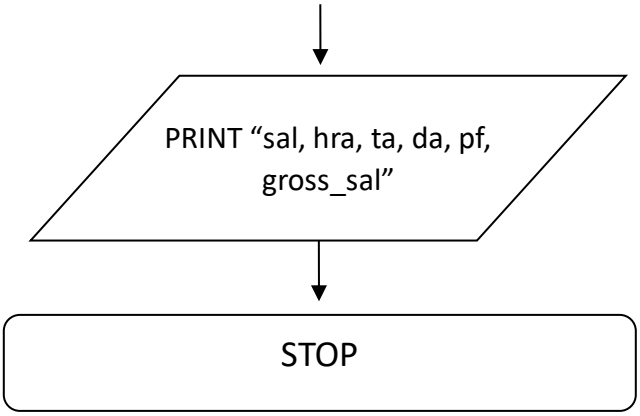
Step 6 : gross_sal = sal + hra + ta + da - pf

Step 7 : PRINT sal, hra, ta, da, pf, gross_sal

Step 8 : STOP

Flowchart :





Code :

```
#include <stdio.h>
#include <conio.h>

void main()
{
   float sal, hra, ta, da, pf, gross_sal;

   printf("\nEnter Salary: ");
   scanf("%f", &sal);

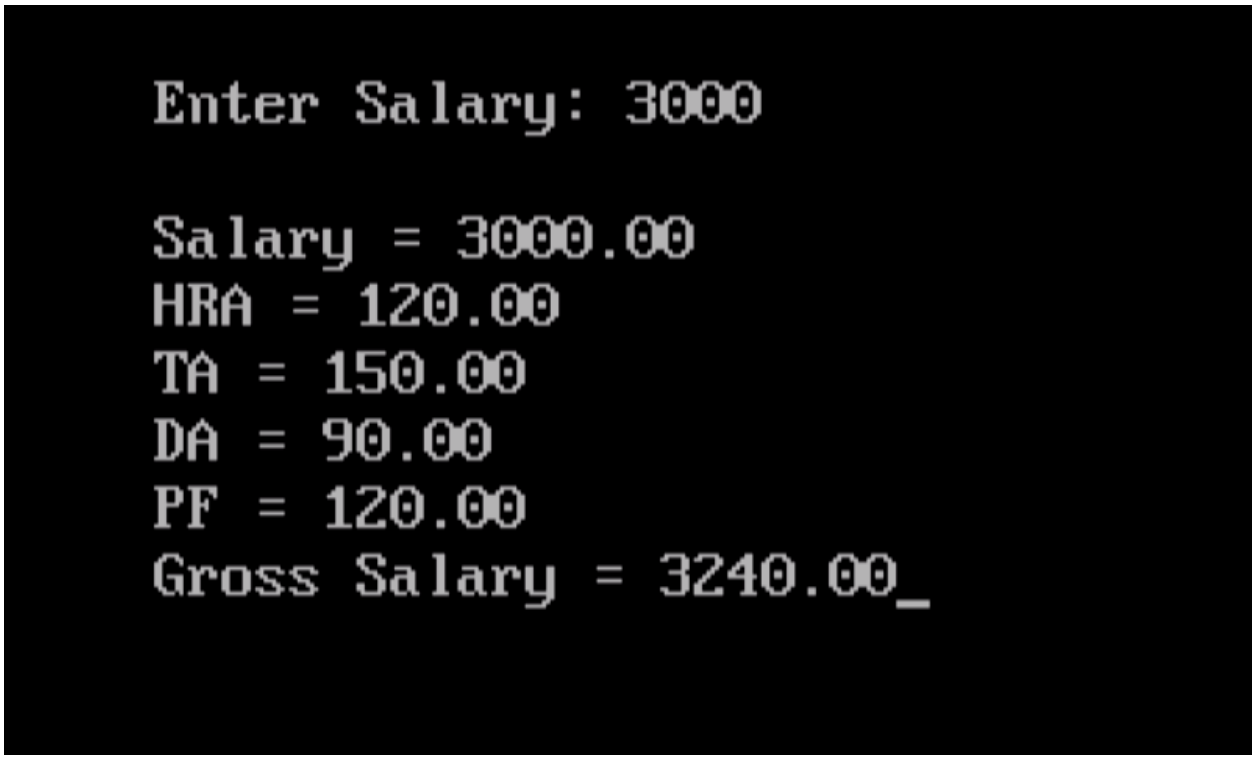
   if (sal >= 5000){
      hra = (5.0 / 100) * sal;
      ta = (6.0 / 100) * sal;
      da = (4.0 / 100) * sal;
```

```
        pf = (5.0 / 100) * sal;
    }
    else{
        hra = (4.0 / 100) * sal;
        ta = (5.0 / 100) * sal;
        da = (3.0 / 100) * sal;
        pf = (4.0 / 100) * sal;
    }
    gross_sal = sal + hra + ta + da - pf;

    printf("\nSalary = %.2f", sal);
    printf("\nHRA = %.2f", hra);
    printf("\nTA = %.2f", ta);
    printf("\nDA = %.2f", da);
    printf("\nPF = %.2f", pf);
    printf("\nGross Salary = %.2f", gross_sal);

    getch();
}
```

Output :


Enter Salary: 3000
Salary = 3000.00
HRA = 120.00
TA = 150.00
DA = 90.00
PF = 120.00
Gross Salary = 3240.00_

(7) Write a C Program to input price quantity of an item from user and display final payment according to following condition. If total purchase(price*quantity) is greater than Rs.1000 then calculate discount 15% otherwise calculate discount 10% of total rupees.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables price, quantity, total, discount, finalAmount

Step 3 : INPUT price, quantity

Step 4 : total = price * quantity

Step 5 : IF total > 1000 THEN

discount = total * 15 / 100

Step 6 : ELSE

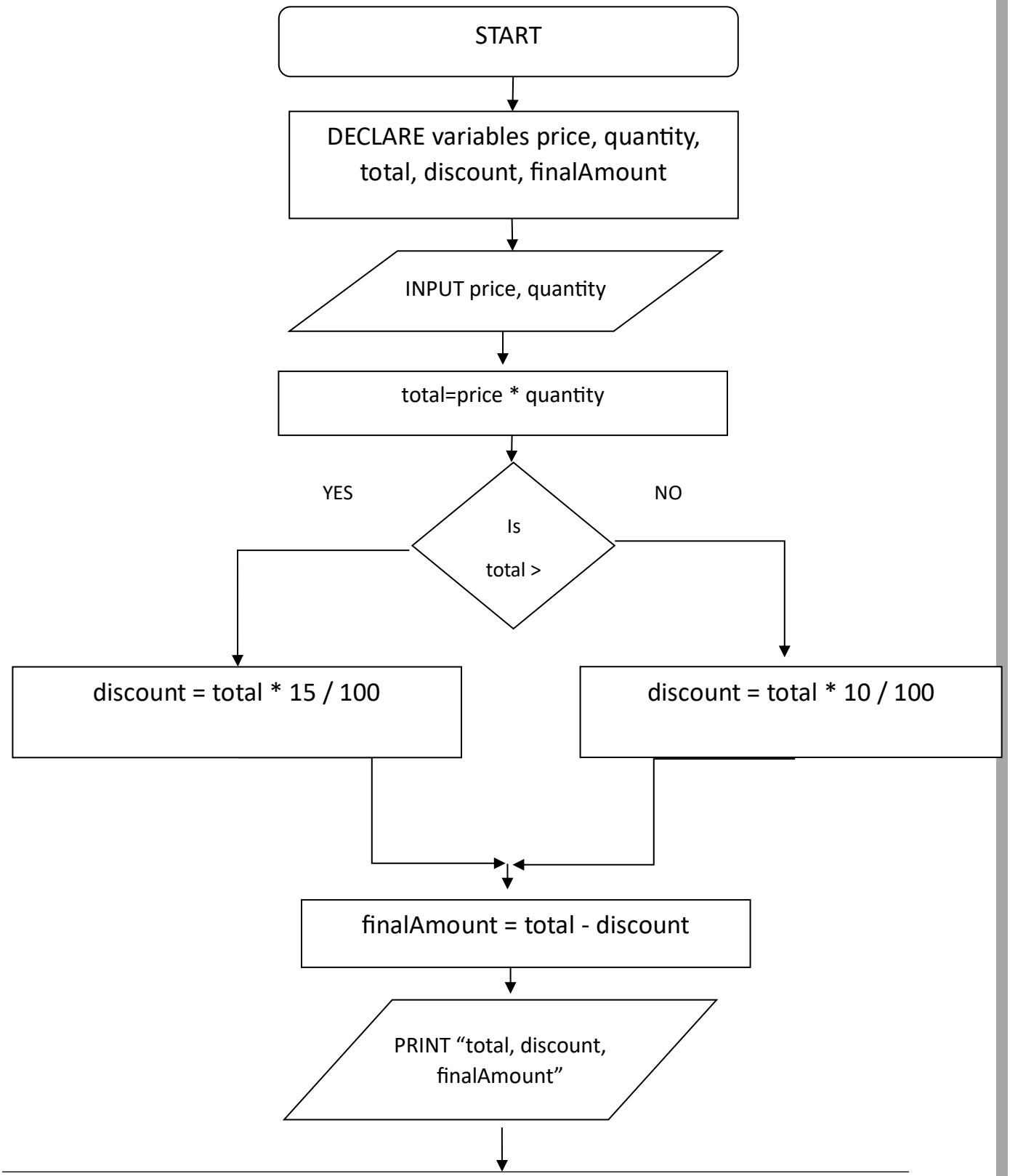
discount = total * 10 / 100

Step 7 : finalAmount = total - discount

Step 8 : PRINT total, discount, finalAmount

Step 9 : STOP

Flowchart :


START
DECLARE variables price, quantity, total, discount, finalAmount
INPUT price, quantity
total=price * quantity
YES
Is total >
NO
discount = total * 15 / 100
discount = total * 10 / 100
finalAmount = total - discount
PRINT “total, discount, finalAmount”

STOP

Code :

```
#include <stdio.h>
#include <conio.h>

int main()
{
   int quantity;
   float price, total, discount, finalAmount;

   clrscr();

   printf("Enter price of item: ");
   scanf("%f", &price);

   printf("Enter quantity: ");
   scanf("%d", &quantity);

   total = price * quantity;
```

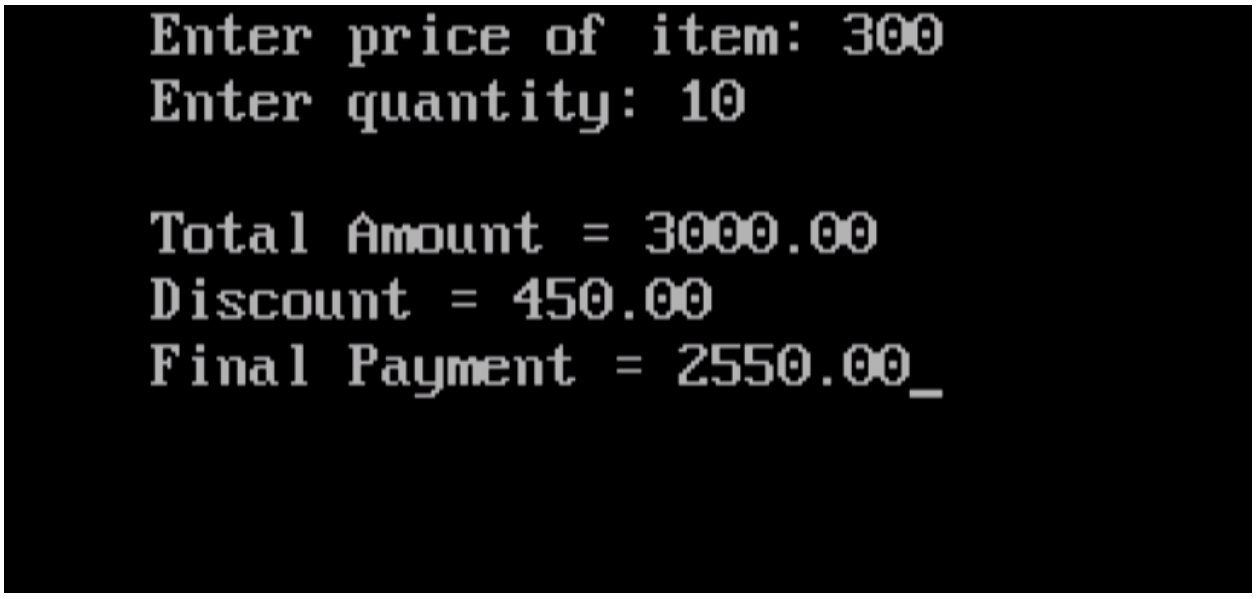
```
    if (total > 1000)
        discount = total * 15 / 100;
    else
        discount = total * 10 / 100;

    finalAmount = total - discount;

    printf("\nTotal Amount = %.2f", total);
    printf("\nDiscount = %.2f", discount);
    printf("\nFinal Payment = %.2f", finalAmount);

    getch();
    return 0;
}
```

Output:


Enter price of item: 300
Enter quantity: 10
Total Amount = 3000.00
Discount = 450.00
Final Payment = 2550.00_

(8) Write a program to input year and find whether year is leap year or not.(Hint Use % Sign)

Algorithm :

Step 1 : START

Step 2 : DECLARE variable year=0

Step 3 : INPUT year

Step 4 : IF (year % 400 = 0) OR (year % 4 = 0 AND year % 100 != 0) THEN

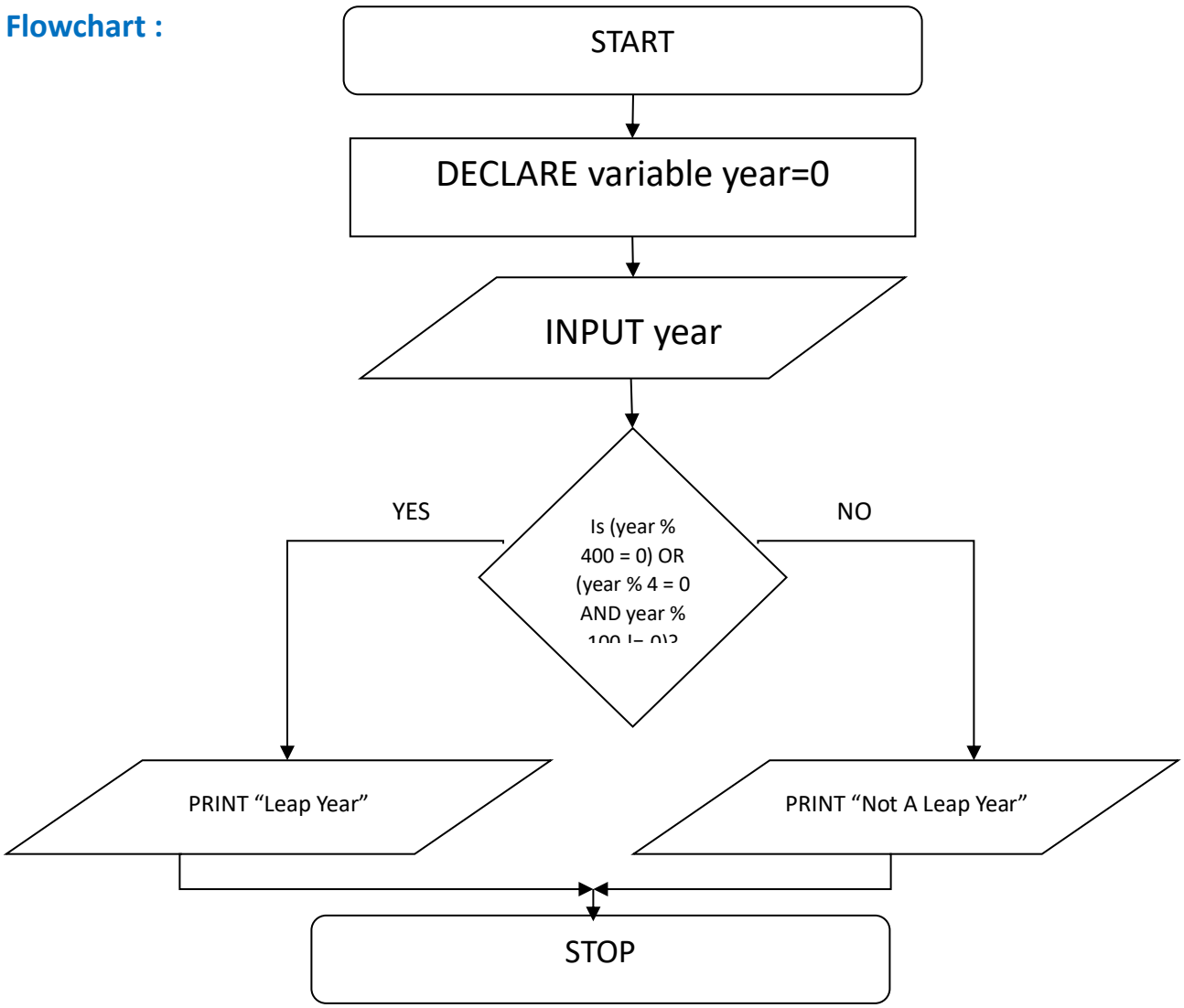
PRINT "Leap Year"

Step 5 : ELSE

PRINT "Not a Leap Year"

Step 6 : STOP

Flowchart :


START
DECLARE variable year=0
INPUT year
Is (year % 400 = 0) OR (year % 4 = 0 AND year %
YES
NO
PRINT “Leap Year”
PRINT “Not A Leap Year”
STOP

Code :

```
#include <stdio.h>
#include <conio.h>

int main()
{
   int year;

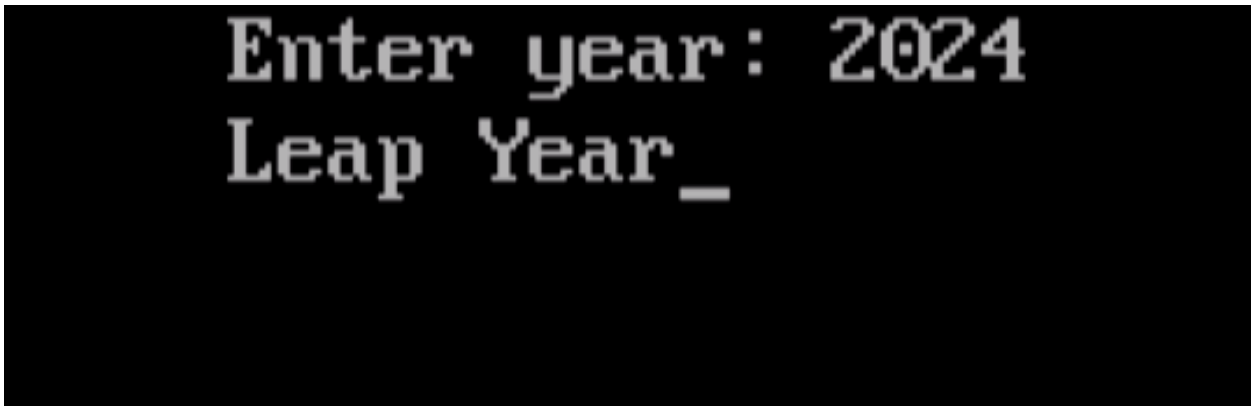
   clrscr();

   printf("Enter year: ");
   scanf("%d", &year);

   if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
      printf("Leap Year");
   else
      printf("Not a Leap Year");

   getch();
   return 0;
}
```

Output :



(9) Write a program to input two values from user and check whether the first number is divisible by second or not. (Hint Use % Sign)

Algorithm :

Step 1 : START

Step 2 : DECLARE variables a=0, b=0

Step 3 : INPUT a, b

Step 4 : IF a % b == 0 THEN

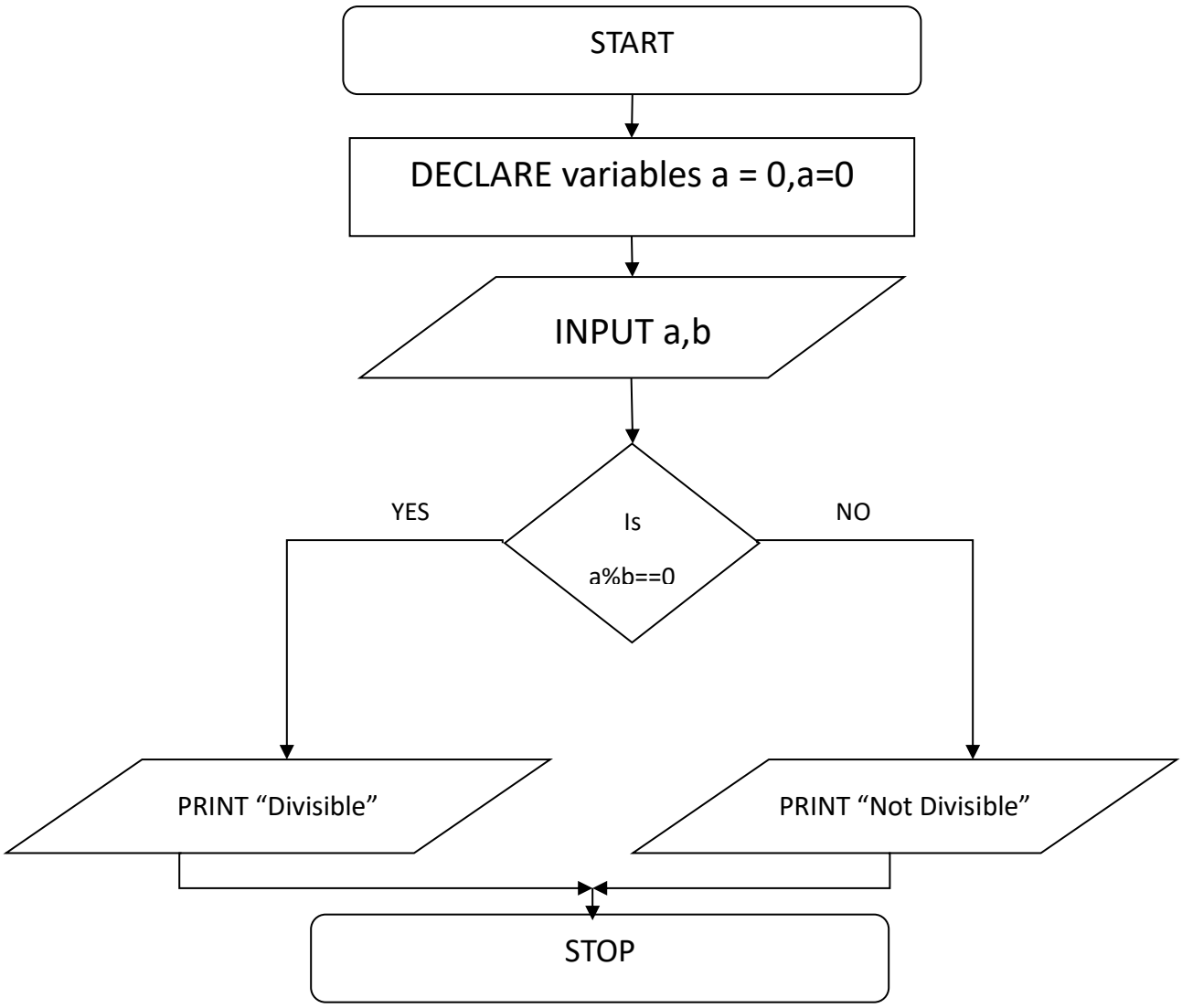
PRINT "a is divisible by b"

Step 5 : ELSE

PRINT "a is not divisible by b"

Step 6 : STOP

Flowchart :


START
DECLARE variables a = 0,a=0
INPUT a,b
Is
a%b==0
YES
NO
PRINT “Divisible”
PRINT “Not Divisible”
STOP

Code :

```
#include <stdio.h>
#include <conio.h>

int main()
{
   int a, b;

   clrscr();

   printf("Enter first number: ");
   scanf("%d", &a);

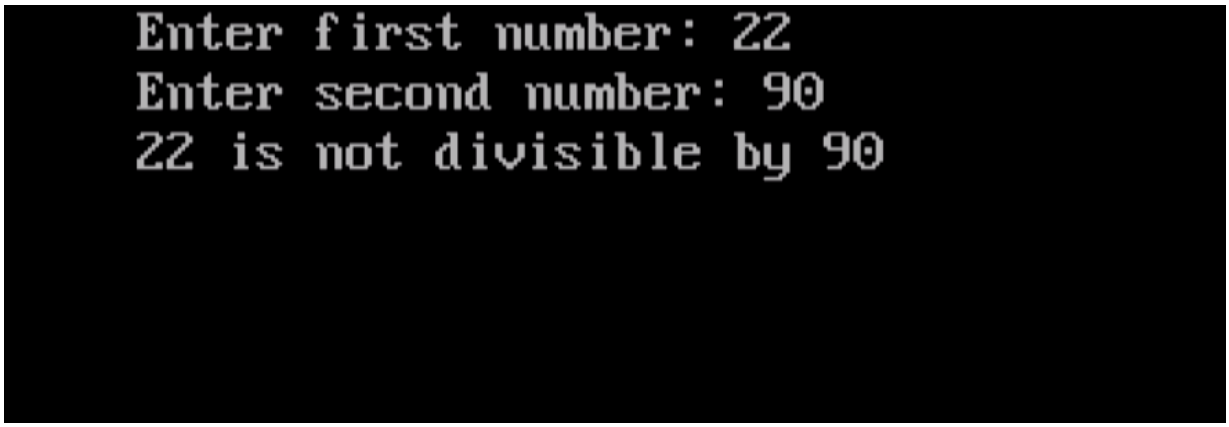
   printf("Enter second number: ");
   scanf("%d", &b);

   if (a % b == 0)
      printf("%d is divisible by %d", a, b);
   else
```

```
        printf("%d is not divisible by %d", a, b);


    getch();
    return 0;
}
```

Output :



(10) Write a program to Input a character, if it is capital convert into small else convert into capital.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable ch

Step 3 : INPUT ch

Step 4 : IF ch >= "A" AND ch <= "Z" THEN

ch = ch + 32

Step 5 : ELSE IF ch >= "a" AND ch <= "z" THEN

ch = ch – 32

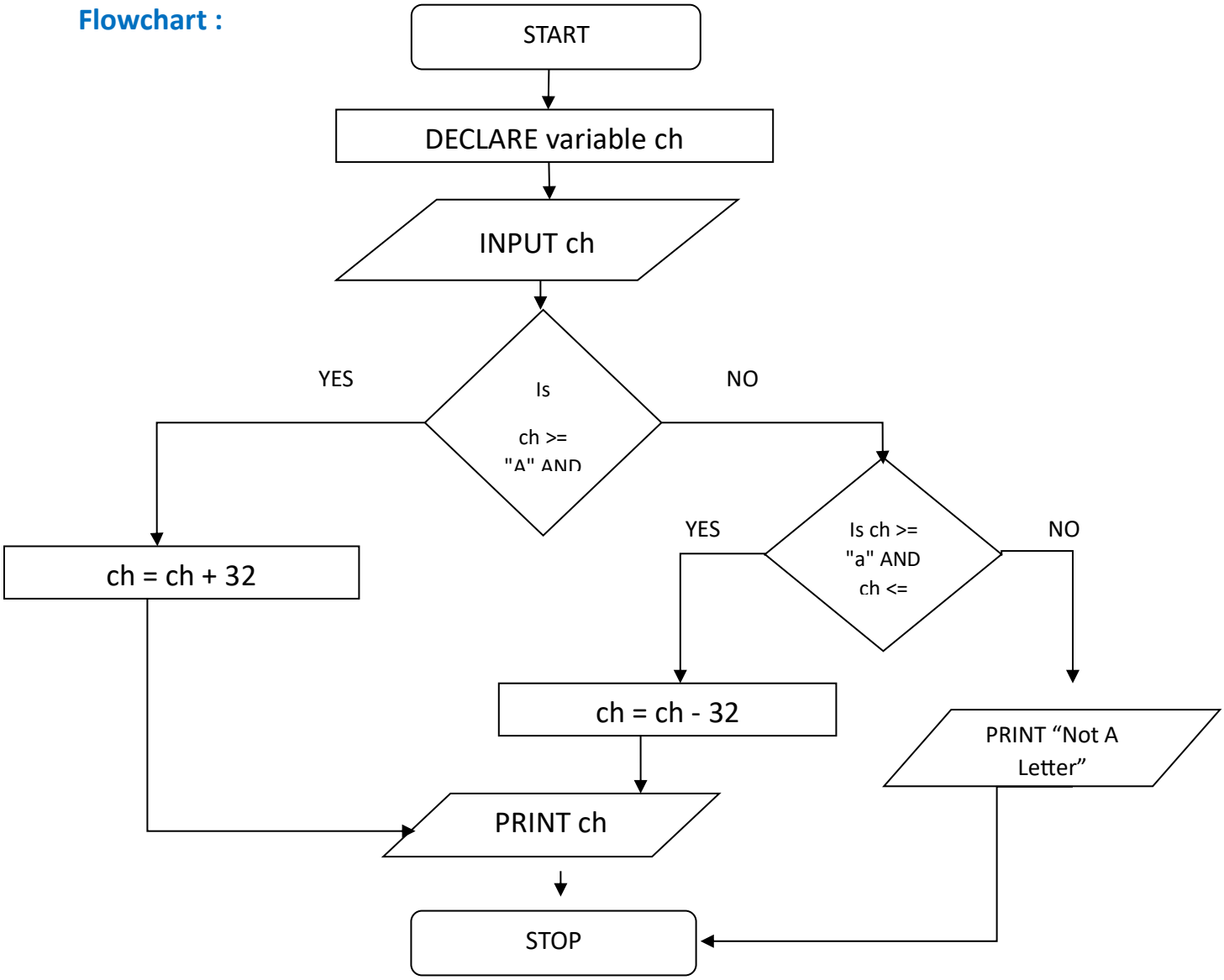
Step 6 : ELSE

PRINT “Not A Letter”

Step 7 : PRINT ch

Step 8 : STOP

Flowchart :


START
DECLARE variable ch
INPUT ch
Is
ch >=
"A" AND
YES
NO
ch = ch + 32
Is ch >=
"a" AND
ch <=
YES
NO
ch = ch - 32
PRINT “Not A
Letter”
PRINT ch
STOP

Code :

```
#include <stdio.h>
#include <conio.h>

int main()
{
   char ch;

   clrscr();

   printf("Enter a character: ");
   scanf(" %c", &ch);

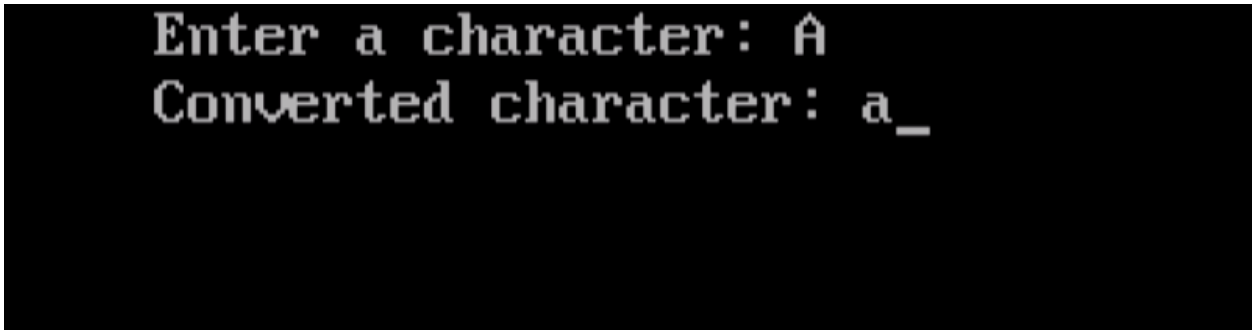
   if (ch >= 'A' && ch <= 'Z')
      ch = ch + 32;
   else if (ch >= 'a' && ch <= 'z')
      ch = ch - 32;

   printf("Converted character: %c", ch);

   getch();
```

```
    return 0;
}
```

Output :



(11) Write a program that input one character from user and find out that character is vowel or not.(aeiou is called vowel).

Algorithm :

Step 1 : START

Step 2 : DECLARE variable ch

Step 3 : INPUT ch

Step 4 : IF ch is vowel (aeiouAEIOU) THEN

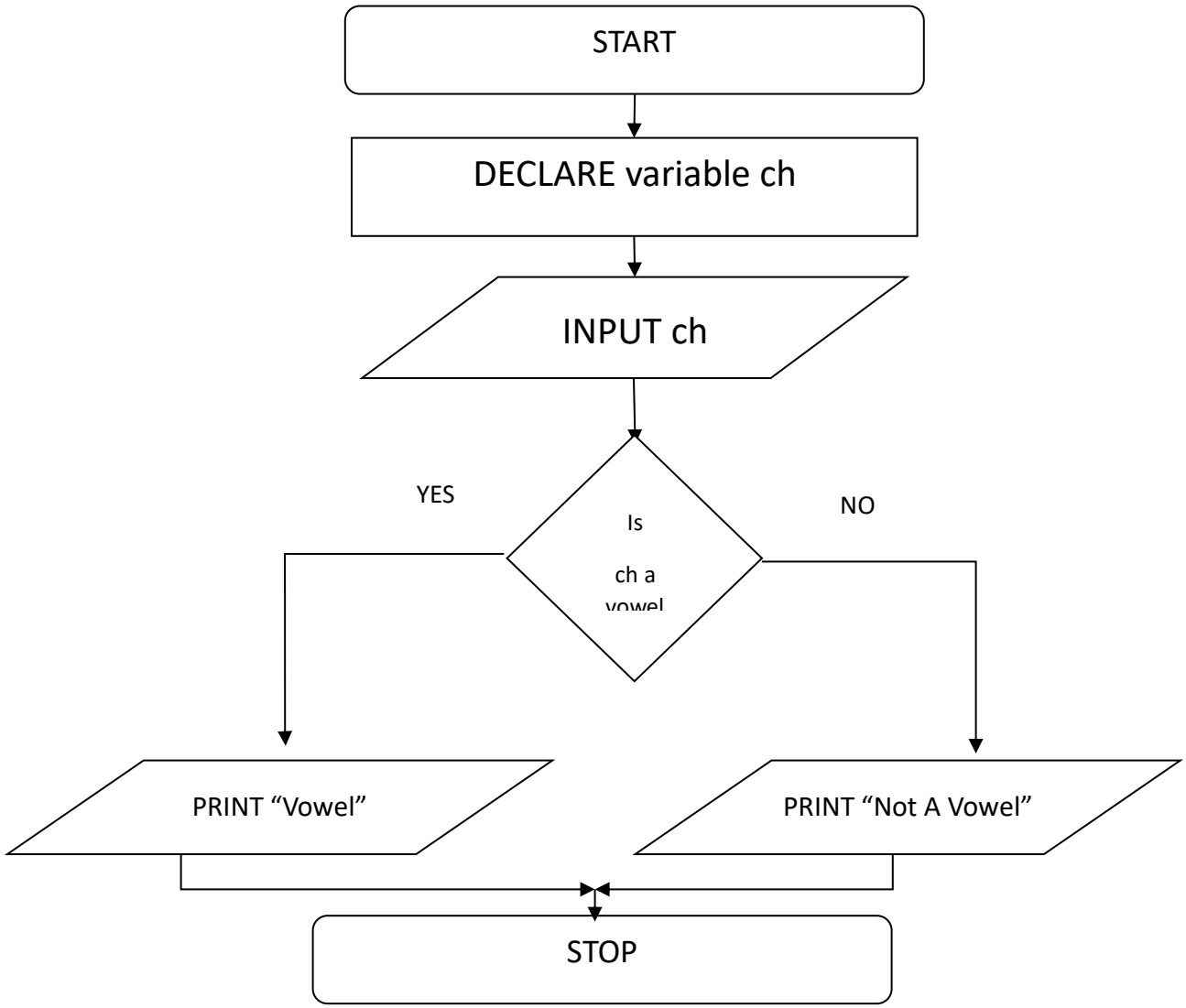
PRINT "Vowel"

Step 5 : ELSE

PRINT "Not a Vowel"

Step 6 : STOP

Flowchart :


START
DECLARE variable ch
INPUT ch
YES
Is
ch a
vowel
NO
PRINT “Vowel”
PRINT “Not A Vowel”
STOP

Code :

```
#include <stdio.h>
#include <conio.h>

int main()
{
   char ch;

   clrscr();

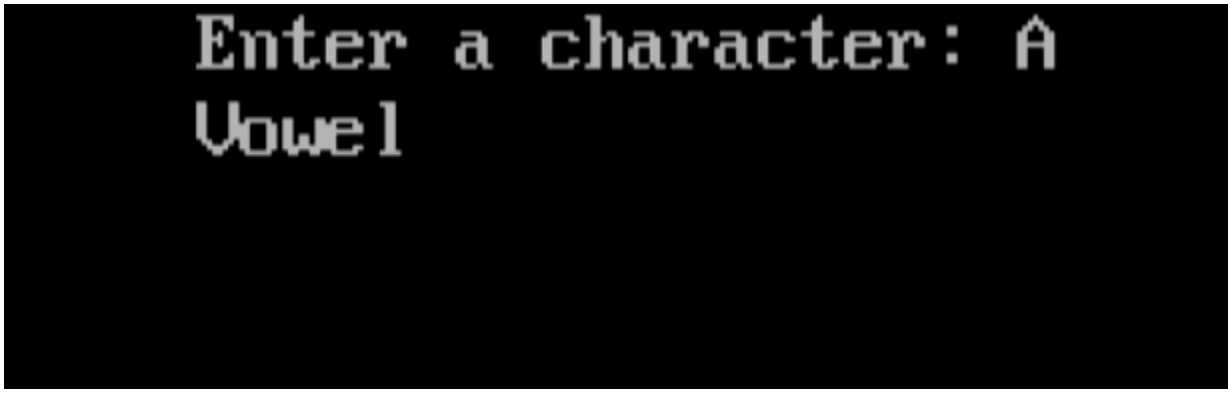
   printf("Enter a character: ");
   scanf(" %c", &ch);

   if (ch=='A'||ch=='E'||ch=='I'||ch=='O'||ch=='U'||
      ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u')
      printf("Vowel");
   else
      printf("Not a Vowel");

   getch();
   return 0;
```

}

Output :



(12) Write programs that find out character is in uppercase or lowercase. Using If…Else..If or Switch.. Case

Algorithm :

Step 1 : START

Step 2 : DECLARE variable ch

Step 3 : INPUT ch

Step 4 : IF ch >= "A" AND ch <= "Z" THEN

PRINT "Uppercase Letter"

Step 5 : ELSE IF ch >= "a" AND ch <= "z" THEN

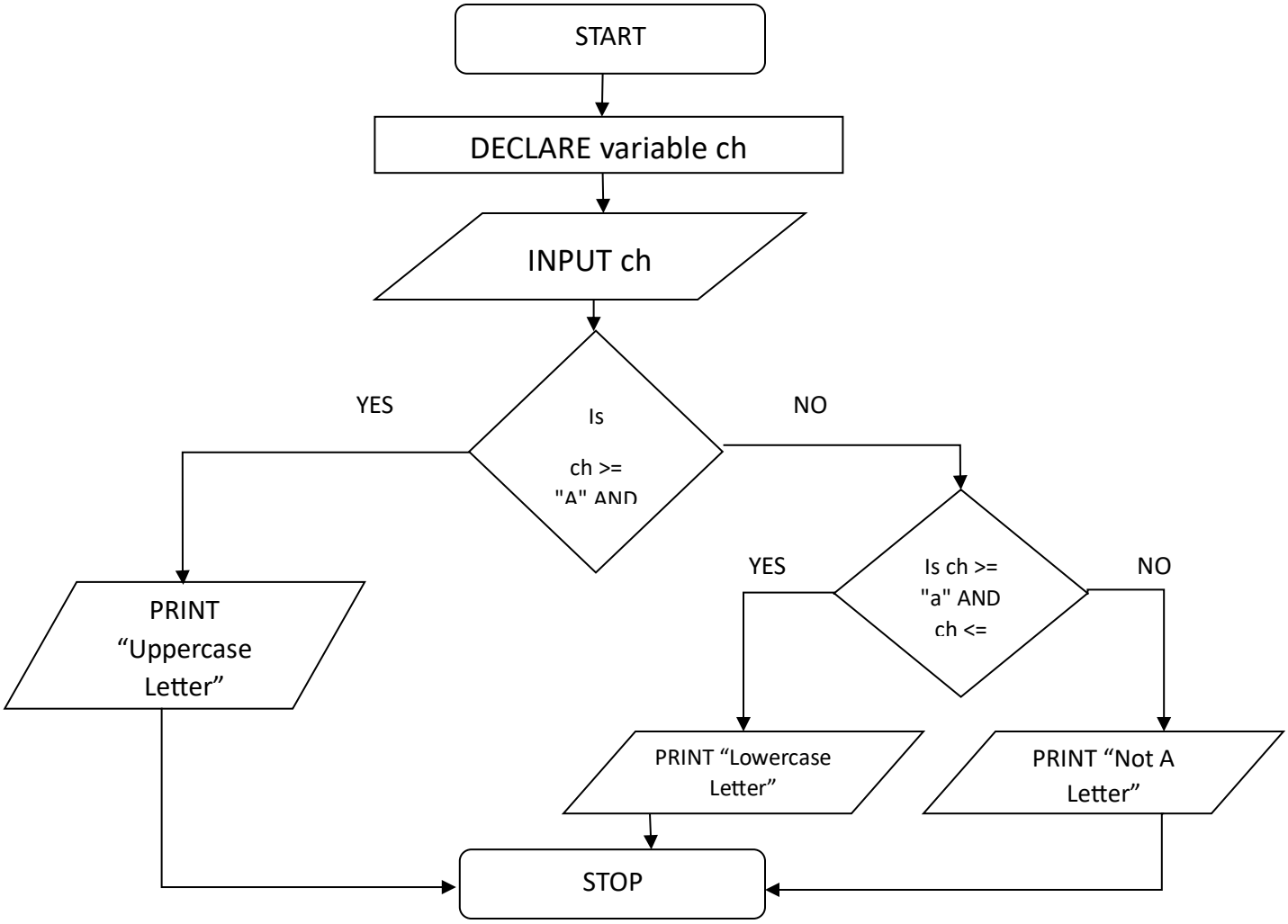
PRINT "Lowercase Letter"

Step 6 : ELSE

PRINT "Not an Alphabet"

Step 7 : STOP

Flowchart :


START
DECLARE variable ch
INPUT ch
Is
ch >=
"A" AND
YES
NO
PRINT
“Uppercase
Letter”
Is ch >=
"a" AND
ch <=
YES
NO
PRINT “Lowercase
Letter”
PRINT “Not A
Letter”
STOP

Code :

```
#include <stdio.h>
#include <conio.h>

int main()
{
   char ch;

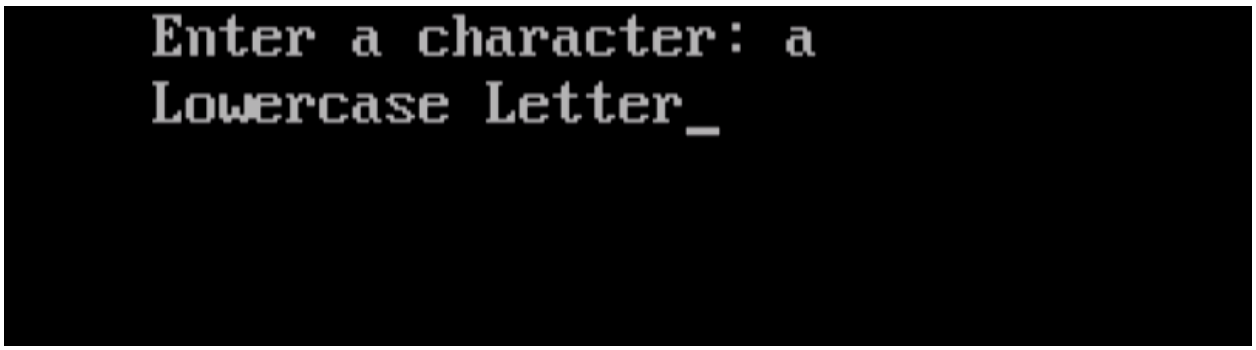
   clrscr();

   printf("Enter a character: ");
   scanf(" %c", &ch);

   if (ch >= 'A' && ch <= 'Z')
      printf("Uppercase Letter");
   else if (ch >= 'a' && ch <= 'z')
      printf("Lowercase Letter");
   else
      printf("Not an Alphabet");
```

```
    getch();
    return 0;
}
```

Output :



(13) Input Three Values from user and Find Maximum.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x, y, z

Step 3 : INPUT x, y, z

Step 4 : IF x > y AND x > z THEN

PRINT "X is Maximum"

Step 5 : ELSE IF y > x AND y > z THEN

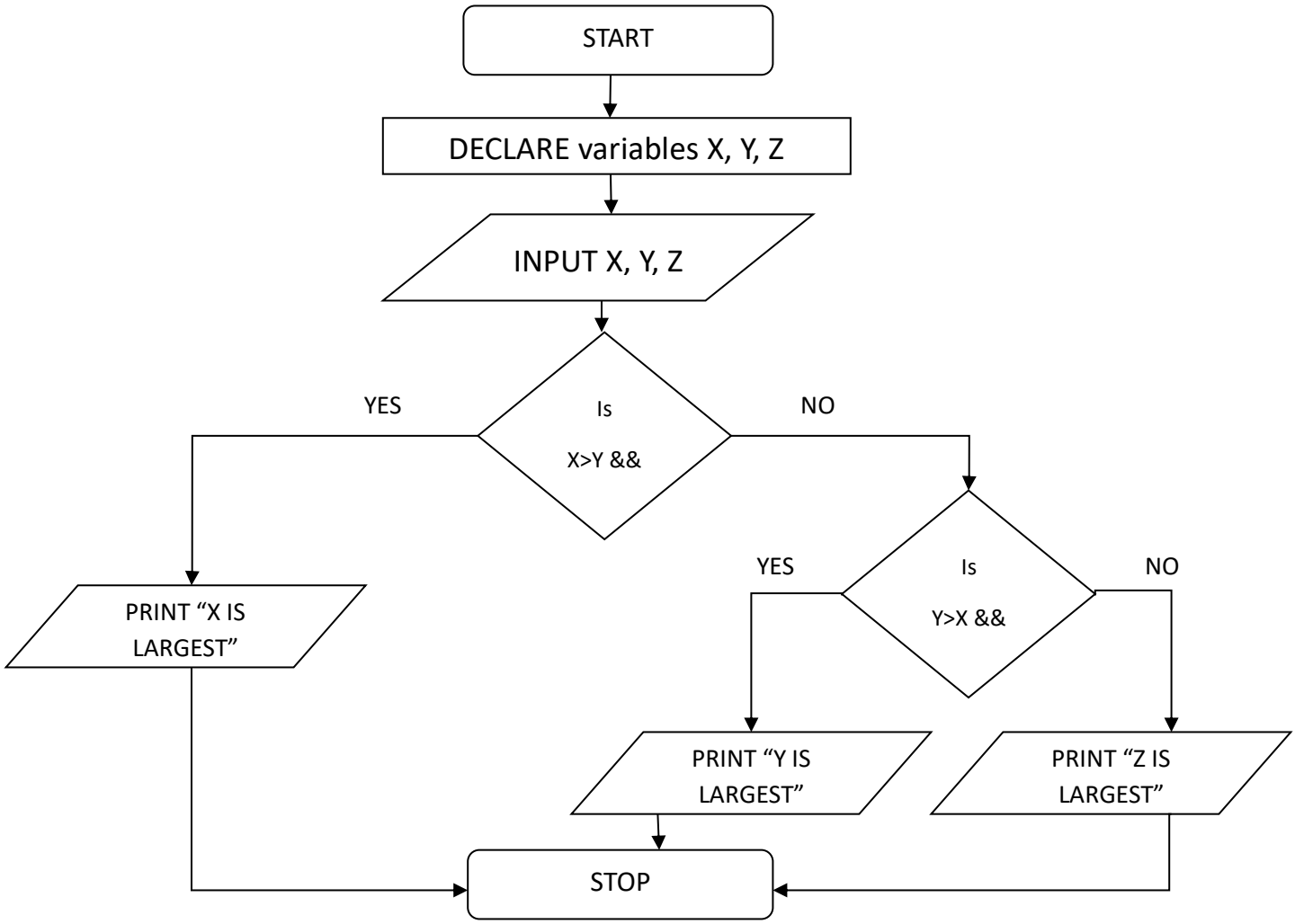
PRINT "Y is Maximum"

Step 6 : ELSE

PRINT "Z is Maximum"

Step 7 : STOP

Flowchart :


START
DECLARE variables X, Y, Z
INPUT X, Y, Z
Is
X>Y &&
YES
NO
PRINT “X IS LARGEST”
Is
Y>X &&
YES
NO
PRINT “Y IS LARGEST”
PRINT “Z IS LARGEST”
STOP

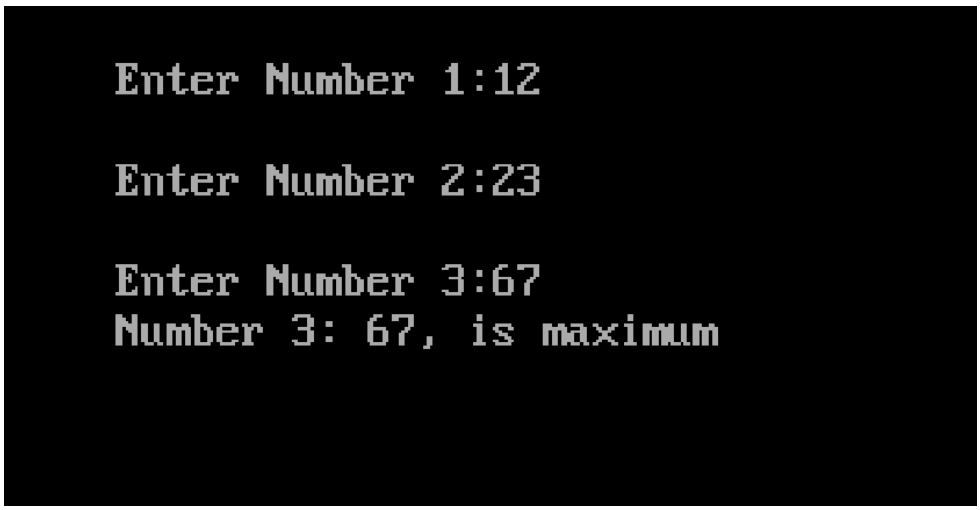
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int x,y,z;
   clrscr();
   printf("\nEnter Number 1:");
   scanf("%d",&x);
   printf("\nEnter Number 2:");
   scanf("%d",&y);
   printf("\nEnter Number 3:");
   scanf("%d",&z);
   if(x>y && x>z){
      printf("Number 1: %d, is maximum", x);
   }
   else if(y>x && y>z){
      printf("Number 2: %d, is maximum", y);
   }
   else
```

```
        printf("Number 3: %d, is maximum", z);
    getch();
}
```

Output :



(14) Input Three Values from user and Find Minimum.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x, y, z

Step 3 : INPUT x, y, z

Step 4 : IF x < y AND x < z THEN

PRINT "X is Minimum"

Step 5 : ELSE IF y < x AND y < z THEN

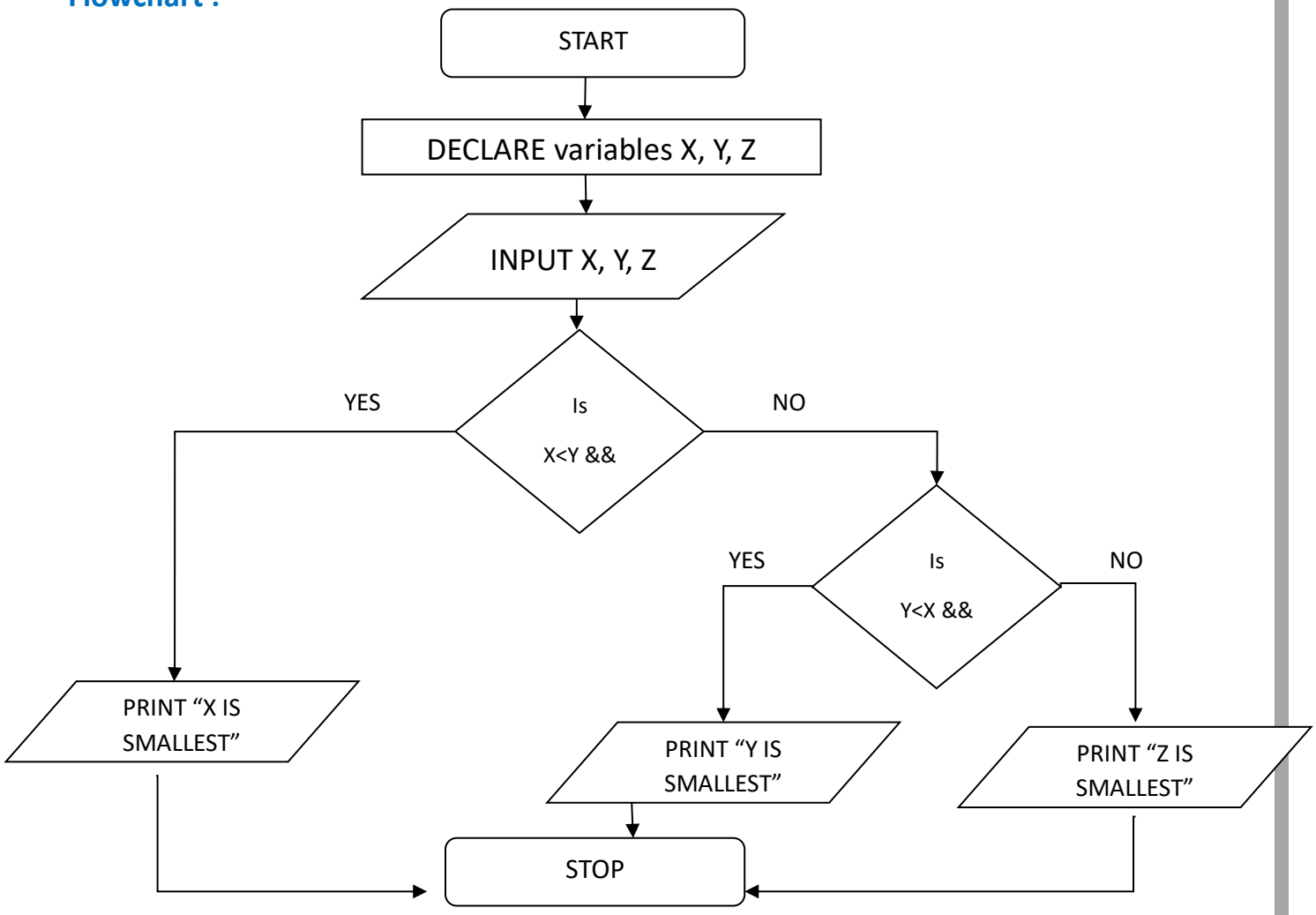
PRINT "Y is Minimum"

Step 6 : ELSE

PRINT "Z is Minimum"

Step 7 : STOP

Flowchart :


START
DECLARE variables X, Y, Z
INPUT X, Y, Z
Is
X<Y &&
YES
NO
Is
Y<X &&
YES
NO
PRINT “X IS SMALLEST”
PRINT “Y IS SMALLEST”
PRINT “Z IS SMALLEST”
STOP

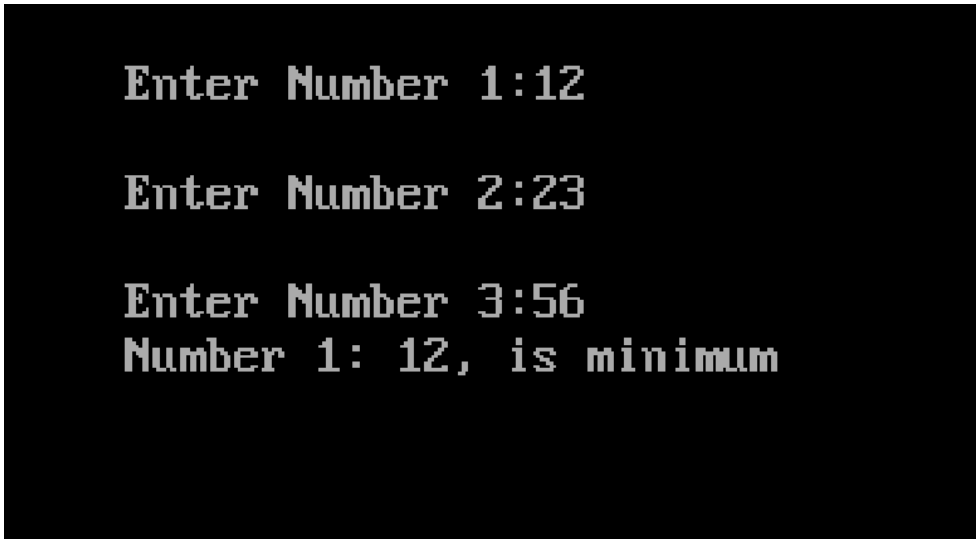
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int x,y,z;
   clrscr();
   printf("\nEnter Number 1:");
   scanf("%d",&x);
   printf("\nEnter Number 2:");
   scanf("%d",&y);
   printf("\nEnter Number 3:");
   scanf("%d",&z);
   if(x<y && x<z){
      printf("Number 1: %d, is minimum", x);
   }
   else if(y<x && y<z){
      printf("Number 2: %d, is minimum", y);
   }
```

```
    else
        printf("Number 3: %d, is minimum", z);
    getch();
}
```

Output :



(15) Input RollNumber, Marks of 5 Subjects and find out Total, Percentage, Result and Grade.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables roll, m1,m2,m3,m4,m5, total, percentage, grade, result

Step 3 : INPUT roll, m1, m2, m3, m4, m5

Step 4 : total = m1 + m2 + m3 + m4 + m5

Step 5 : percentage = total / 5

Step 6 : IF any mark < 35 THEN

result = Fail

grade = F

Step 7 : ELSE

result = Pass

Step 8 : IF percentage >= 90 THEN

grade = A+

Step 9 : ELSE IF percentage >= 80 THEN

grade = A

Step 10 : ELSE IF percentage >= 70 THEN

grade = B

Step 11 : ELSE IF percentage >= 60 THEN

grade = C

Step 12 : ELSE IF percentage >= 50 THEN

grade = D

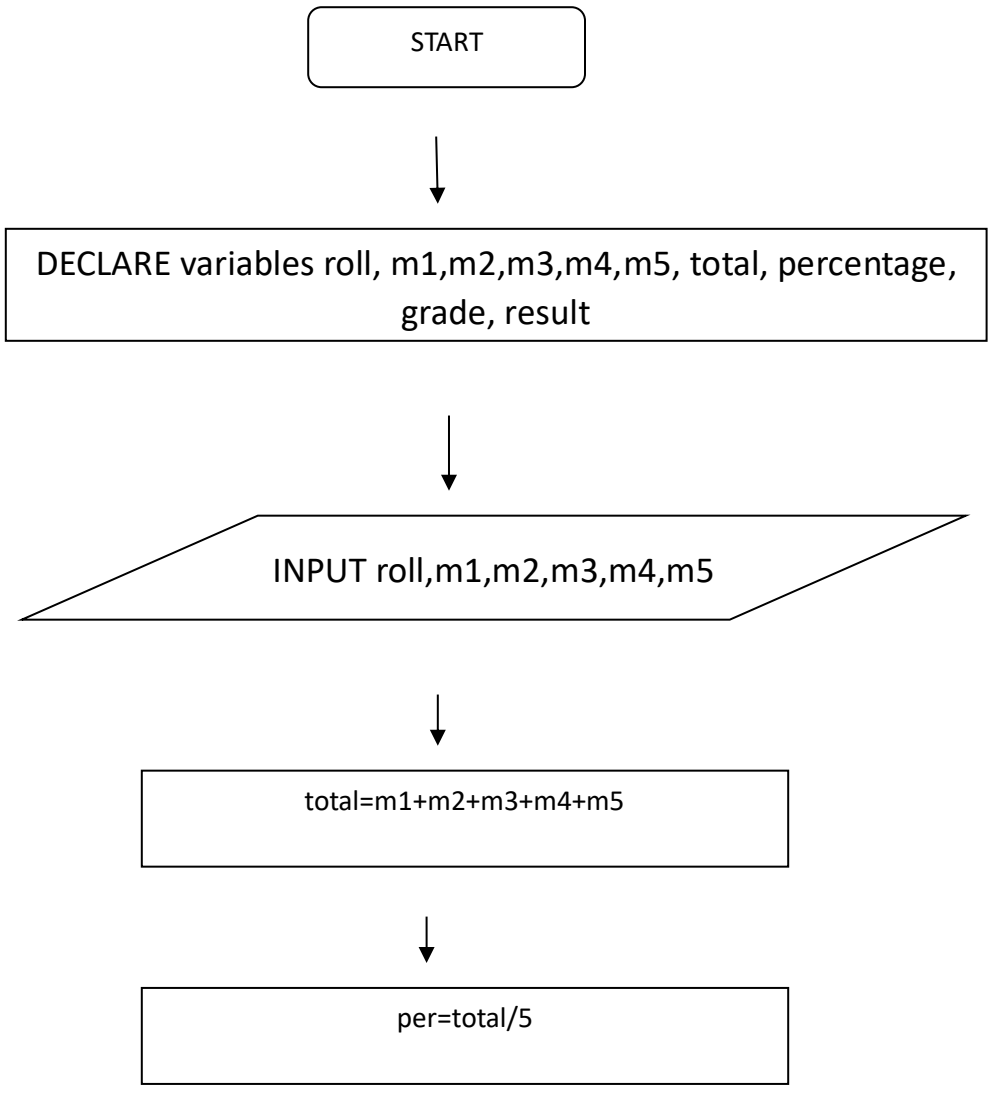
Step 13 : ELSE

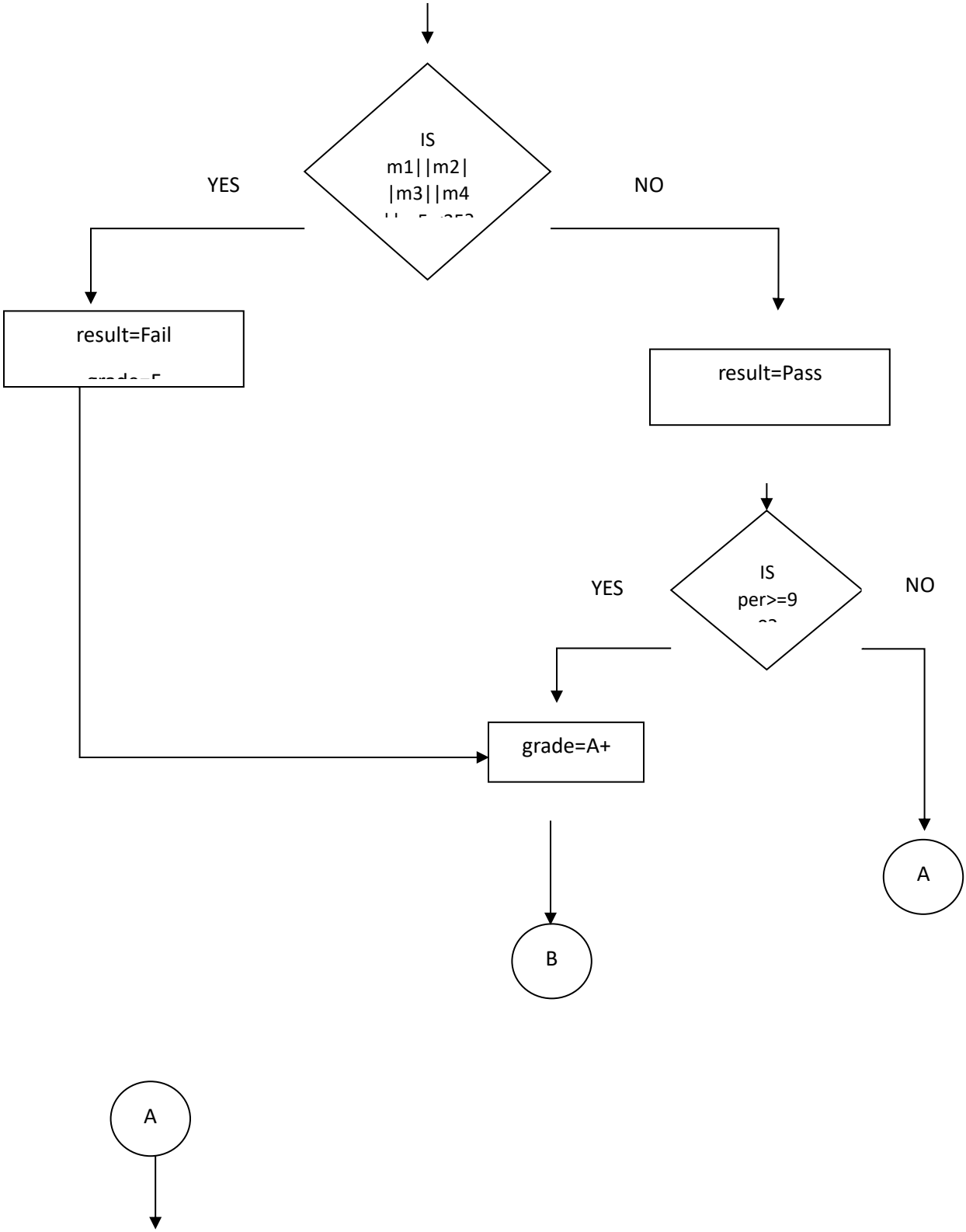
grade = E

Step 14 : PRINT total, percentage, grade, result

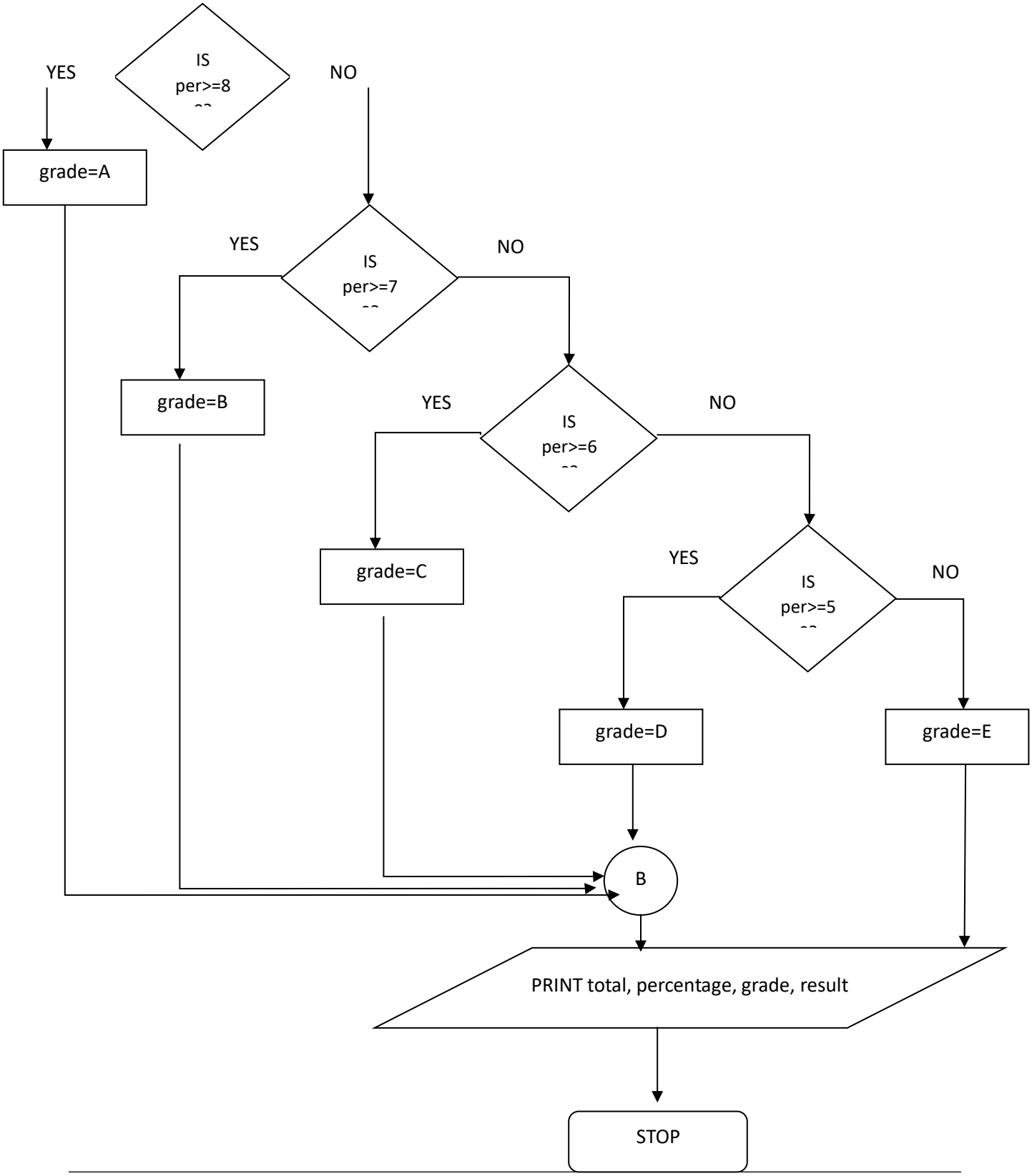
Step 15 : STOP

Flowchart :


START
DECLARE variables roll, m1,m2,m3,m4,m5, total, percentage, grade, result
INPUT roll,m1,m2,m3,m4,m5
total=m1+m2+m3+m4+m5
per=total/5



Roll no.92600527009 NAME:TIRTH BHIMANI



Roll no.92600527009 NAME:TIRTH BHIMANI

Code :

```
#include <stdio.h>
#include <conio.h>

int main()
{
   int roll, m1, m2, m3, m4, m5;
   int total;
   float percentage;

   clrscr();

   printf("Enter Roll Number: ");
   scanf("%d", &roll);

   printf("Enter marks of 5 subjects:\n");
   scanf("%d%d%d%d%d", &m1, &m2, &m3, &m4, &m5);

   total = m1 + m2 + m3 + m4 + m5;
   percentage = total / 5.0;
```

```
printf("\nRoll Number = %d", roll);
printf("\nTotal = %d", total);
printf("\nPercentage = %.2f", percentage);

if (m1 < 35 || m2 < 35 || m3 < 35 || m4 < 35 || m5 < 35)
{
   printf("\nResult = Fail");
   printf("\nGrade = F");
}
else
{
   printf("\nResult = Pass");

   if (percentage >= 90)
      printf("\nGrade = A+");
   else if (percentage >= 80)
      printf("\nGrade = A");
   else if (percentage >= 70)
      printf("\nGrade = B");
   else if (percentage >= 60)
      printf("\nGrade = C");
   else if (percentage >= 50)
```

```
            printf("\nGrade = D");
        else
            printf("\nGrade = E");
    }

    getch();
    return 0;
}
```

Output :

```
Enter Roll Number: 7044
Enter marks of 5 subjects:
78
88
89
70
87

Roll Number = 7044
Total = 412
Percentage = 82.40
Result = Pass
Grade = A
```

(16) Input day in number and print in text(for Example input 1 then Output Monday)

Algorithm :

Step 1 : START

Step 2 : DECLARE variable day

Step 3 : INPUT day

Step 4 : IF day == 1 THEN

PRINT "Monday"

Step 5 : ELSE IF day == 2 THEN

PRINT "Tuesday"

Step 6 : ELSE IF day == 3 THEN

PRINT "Wednesday"

Step 7 : ELSE IF day == 4 THEN

PRINT "Thursday"

Step 8 : ELSE IF day == 5 THEN

PRINT "Friday"

Step 9 : ELSE IF day == 6 THEN

PRINT "Saturday"

Step 10 : ELSE IF day == 7 THEN

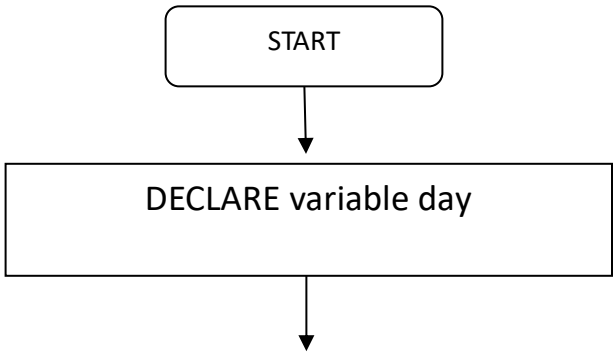
PRINT "Sunday"

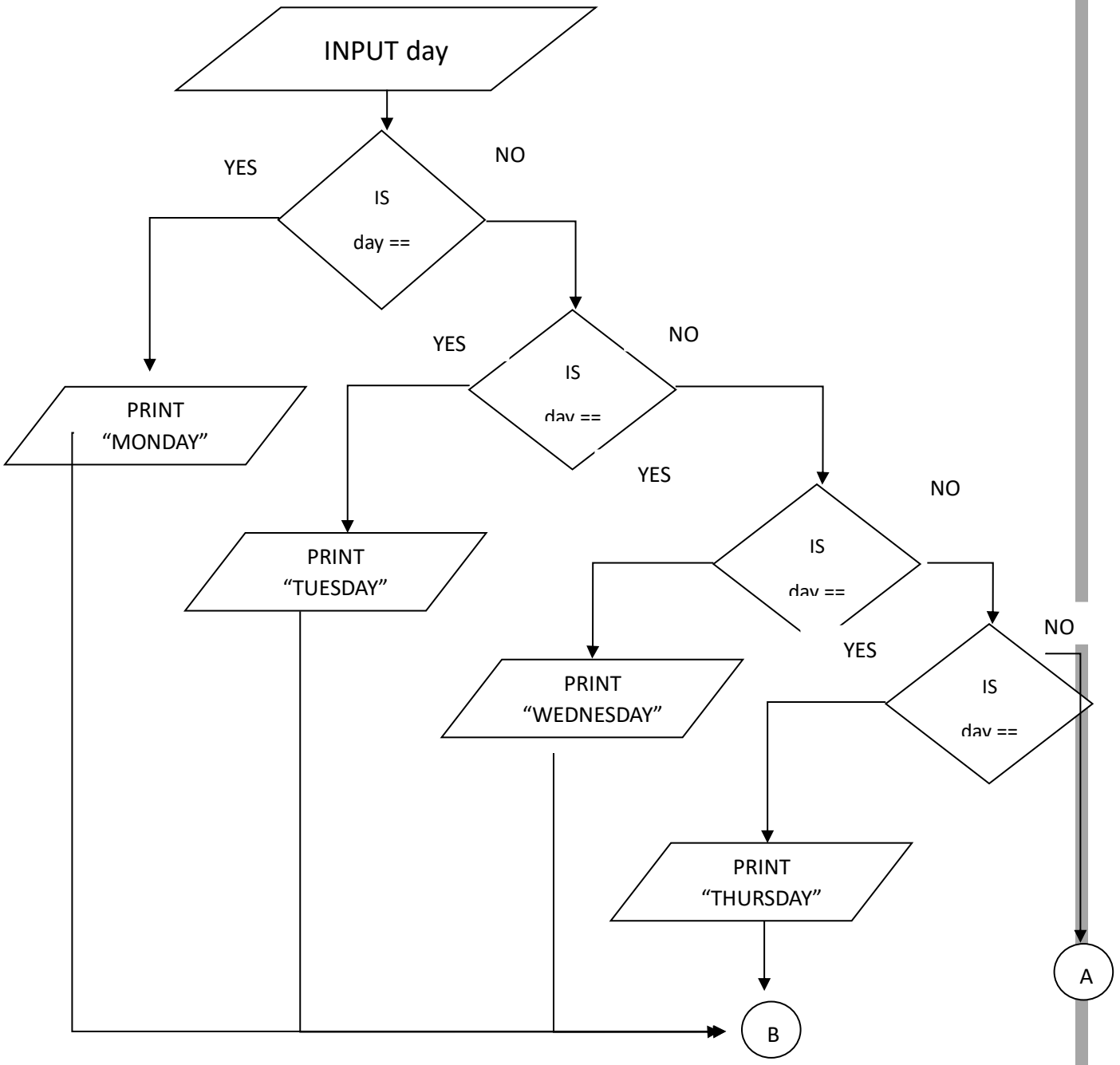
Step 11 : ELSE

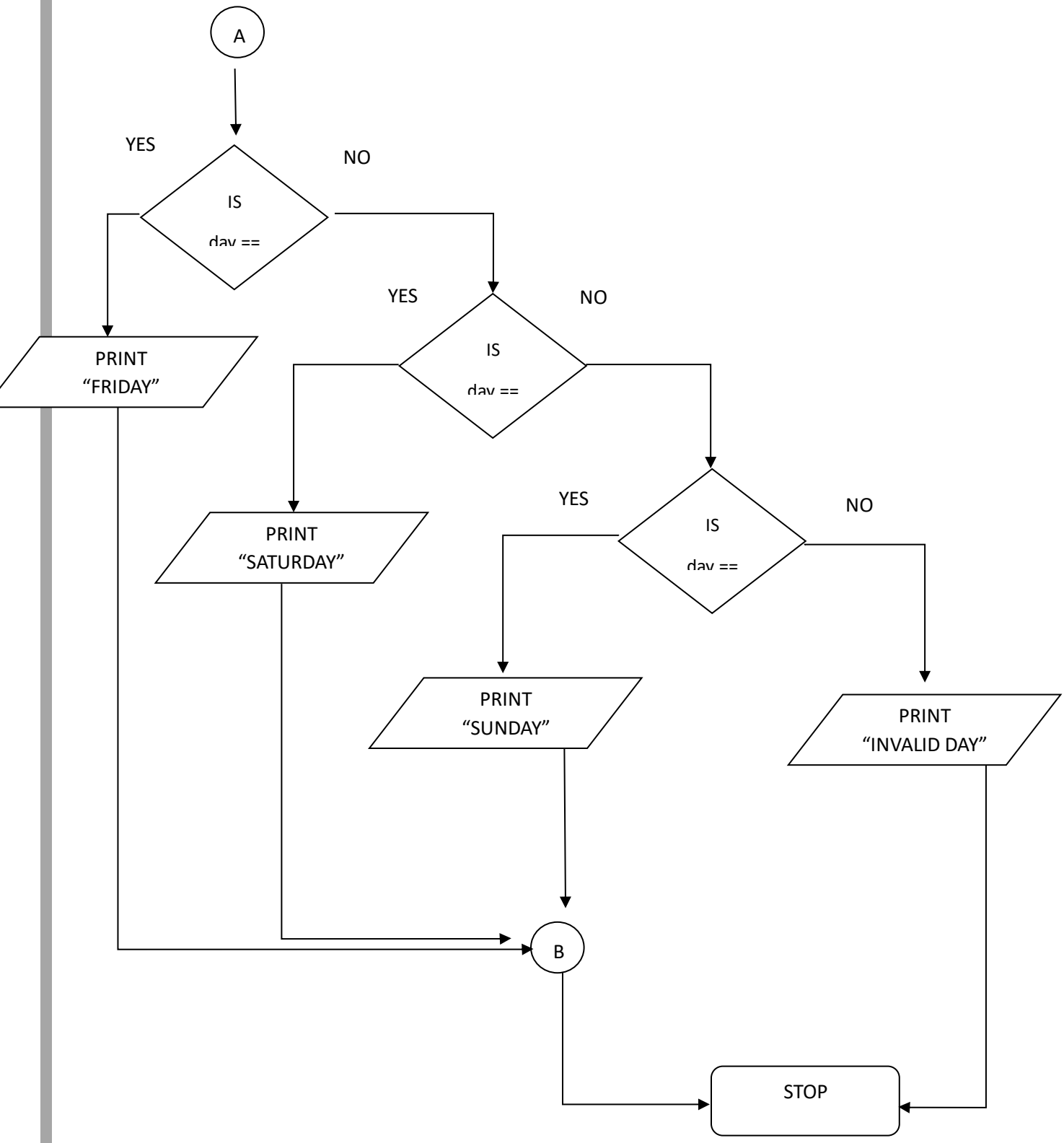
PRINT "Invalid Day"

Step 12 : STOP

Flowchart :




INPUT day
IS
day ==
YES
NO
PRINT
“MONDAY”
IS
day ==
YES
NO
PRINT
“TUESDAY”
IS
day ==
YES
NO
PRINT
“WEDNESDAY”
IS
day ==
YES
NO
PRINT
“THURSDAY”
A
B



Roll no.92600527009 NAME:TIRTH BHIMANI

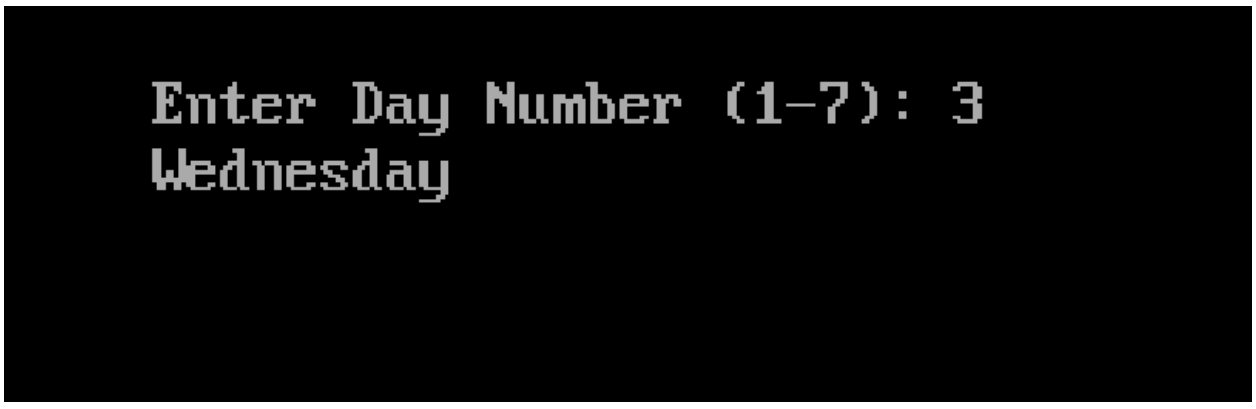
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int i;
   clrscr();
   printf("\nEnter Day Number (1-7): ");
   scanf("%d",&i);
   if(i==1){
      printf("Monday");
   }
   else if(i==2){
      printf("Tuesday");
   }
   else if(i==3){
      printf("Wednesday");
   }
   else if(i==4){
      printf("Thursday");
   }
```

```
    else if(i==5){
        printf("Friday");
    }
    else if(i==6){
        printf("Saturday");
    }
    else if(i==7){
        printf("Sunday");
    }
    else{
        printf("Invalid Day");
    }
    getch();
}
```

Output :



(17) Input Month in number and print in text(for example input 1 then output January)

Algorithm :

Step 1 : START

Step 2 : DECLARE variable month

Step 3 : INPUT month

Step 4 : IF month = 1 THEN

PRINT "January"

Step 5 : ELSE IF month = 2 THEN

PRINT "February"

Step 6 : ELSE IF month = 3 THEN

PRINT "March"

Step 7 : ELSE IF month = 4 THEN

PRINT "April"

Step 8 : ELSE IF month = 5 THEN

PRINT "May"

Step 9 : ELSE IF month = 6 THEN

PRINT "June"

Step 10 : ELSE IF month = 7 THEN

PRINT "July"

Step 11 : ELSE IF month = 8 THEN

PRINT "August"

Step 12 : ELSE IF month = 9 THEN

PRINT "September"

Step 13 : ELSE IF month = 10 THEN

PRINT "October"

Step 14 : ELSE IF month = 11 THEN

PRINT "November"

Step 15 : ELSE IF month = 12 THEN

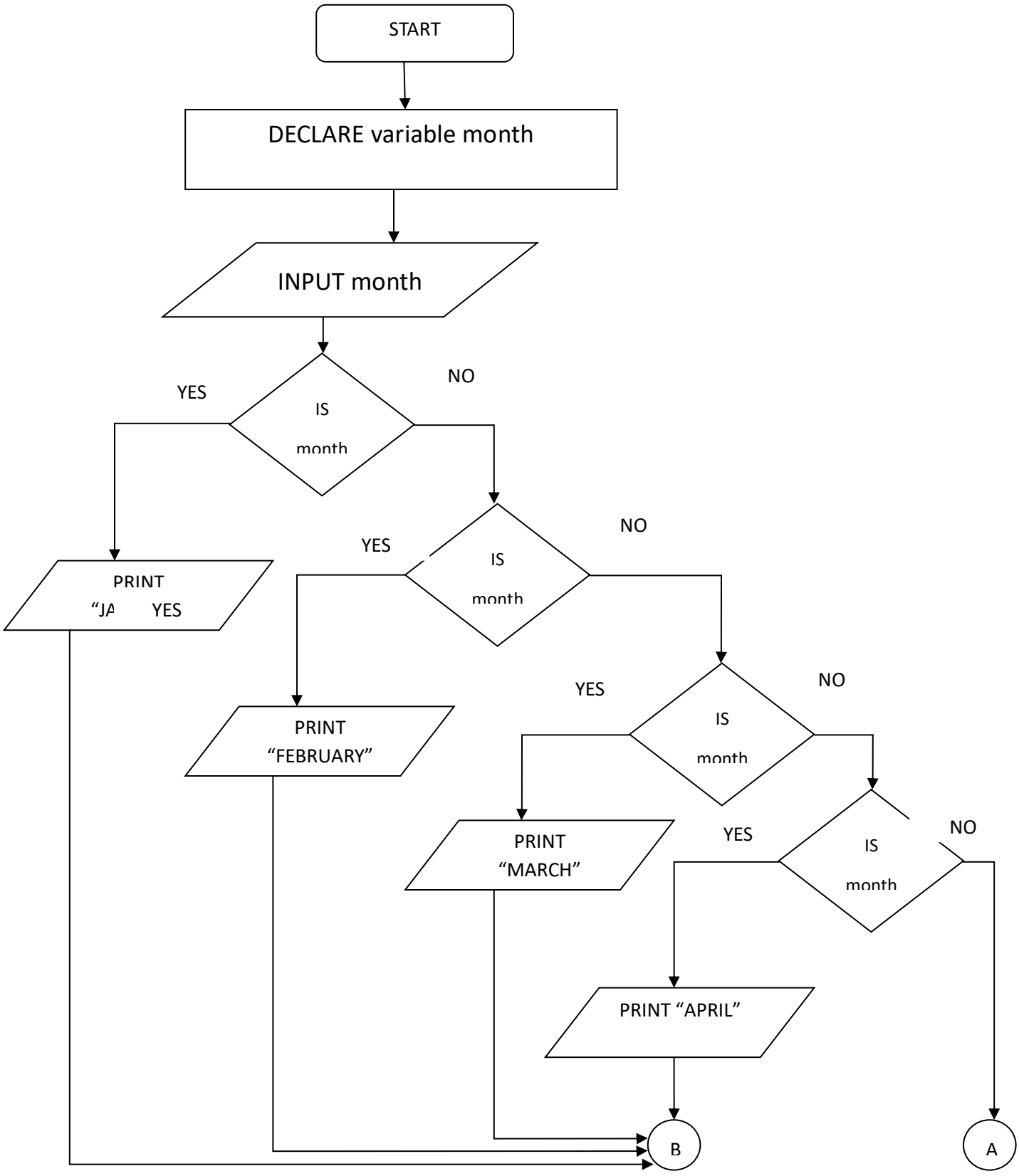
PRINT "December"

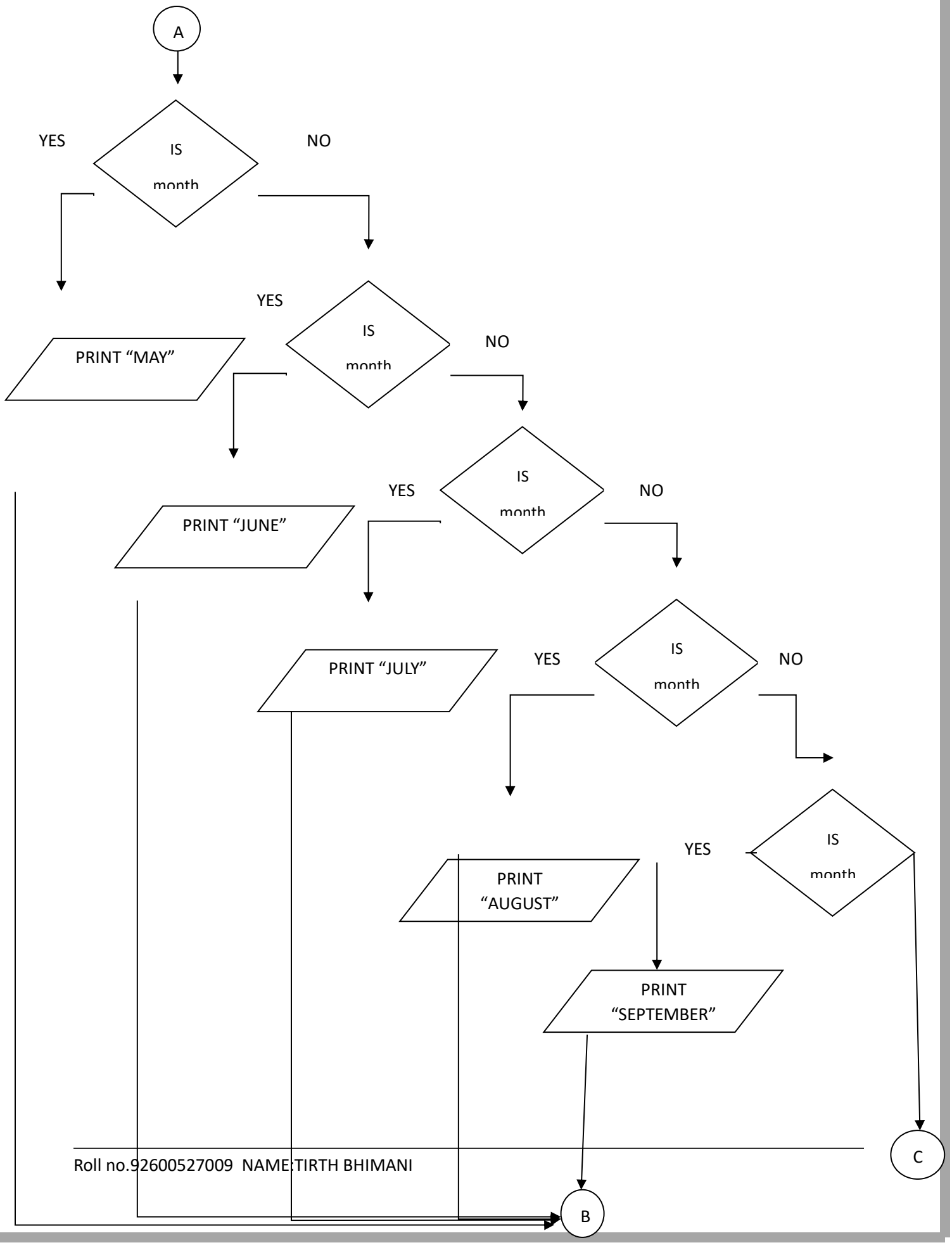
Step 16 : ELSE

PRINT "Invalid Month"

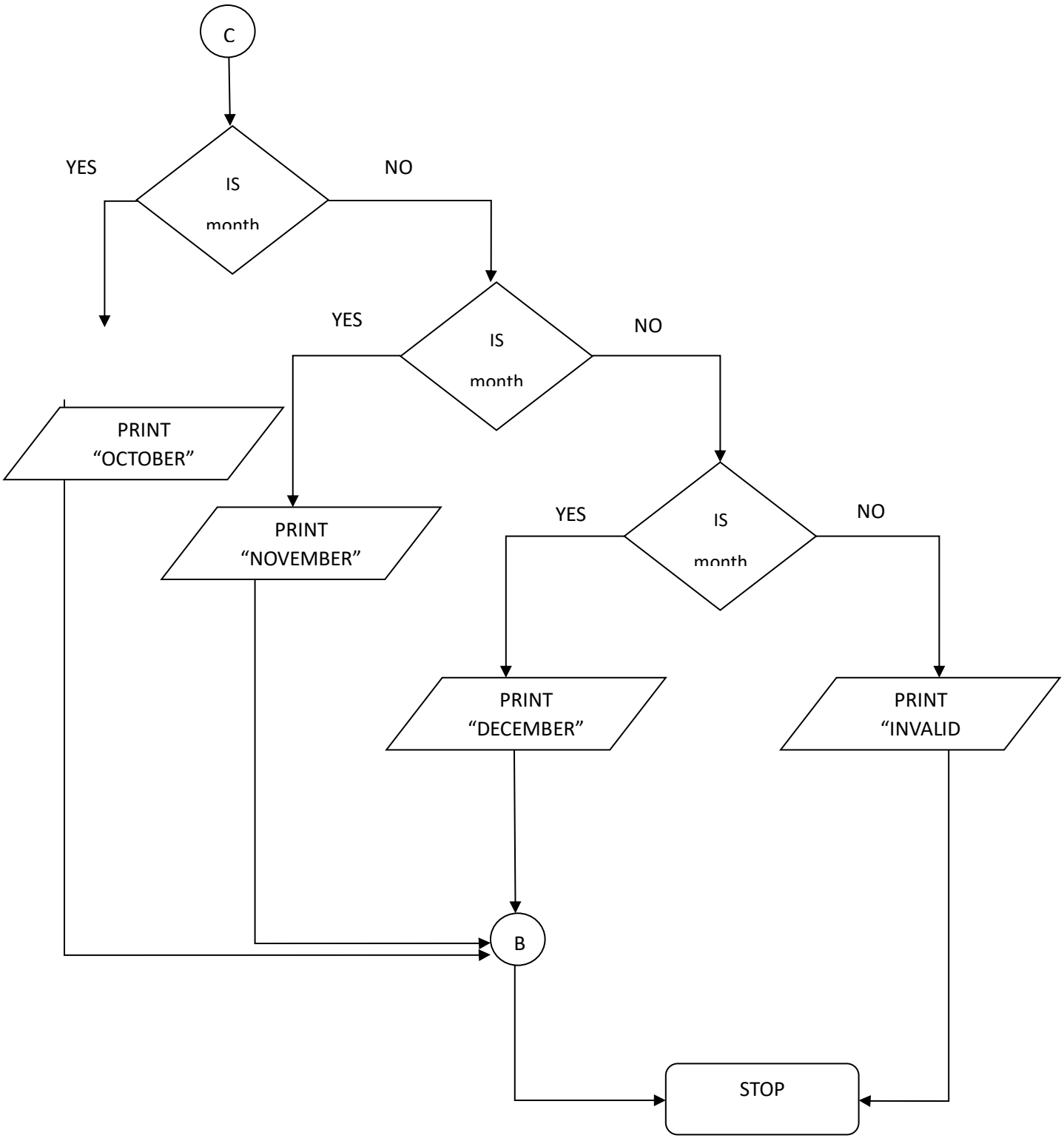
Step 17 : STOP

Flowchart :


START
DECLARE variable month
INPUT month
IS month
YES
NO
PRINT "JA YES
IS month
YES
NO
PRINT "FEBRUARY"
IS month
YES
NO
PRINT "MARCH"
IS month
YES
NO
PRINT "APRIL"
B
A



Roll no.92600527009 NAME:TIRTH BHIMANI


C
YES
IS
month
NO
YES
IS
month
NO
PRINT
“OCTOBER”
PRINT
“NOVEMBER”
YES
IS
month
NO
PRINT
“DECEMBER”
PRINT
“INVALID
B
STOP

Code :

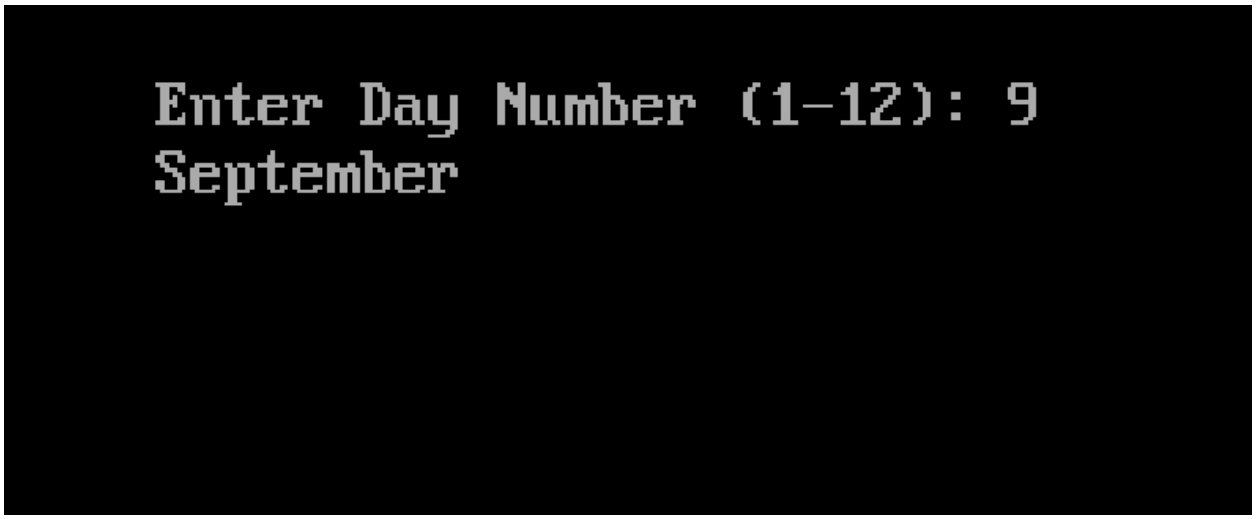
```
#include<stdio.h>
#include<conio.h>

void main(){
   int i;
   clrscr();
   printf("\nEnter Day Number (1-12): ");
   scanf("%d",&i);
   if(i==1){
      printf("January");
   }
   else if(i==2){
      printf("February");
   }
   else if(i==3){
      printf("March");
   }
   else if(i==4){
      printf("April");
   }
```

```
else if(i==5){
   printf("May");
}
else if(i==6){
   printf("June");
}
else if(i==7){
   printf("July");
}
else if(i==8){
   printf("August");
}
else if(i==9){
   printf("September");
}
else if(i==10){
   printf("October");
}
else if(i==11){
   printf("November");
}
else if(i==12){
```

```
        printf("December");
    }
    else{
        printf("Invalid Month");
    }
    getch();
}
```

Output :



(18) Input number and check whether the number is positive, negative or zero.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable x

Step 3 : INPUT x

Step 4 : IF x == 0 THEN

PRINT "Number is Zero"

Step 5 : ELSE IF x > 0 THEN

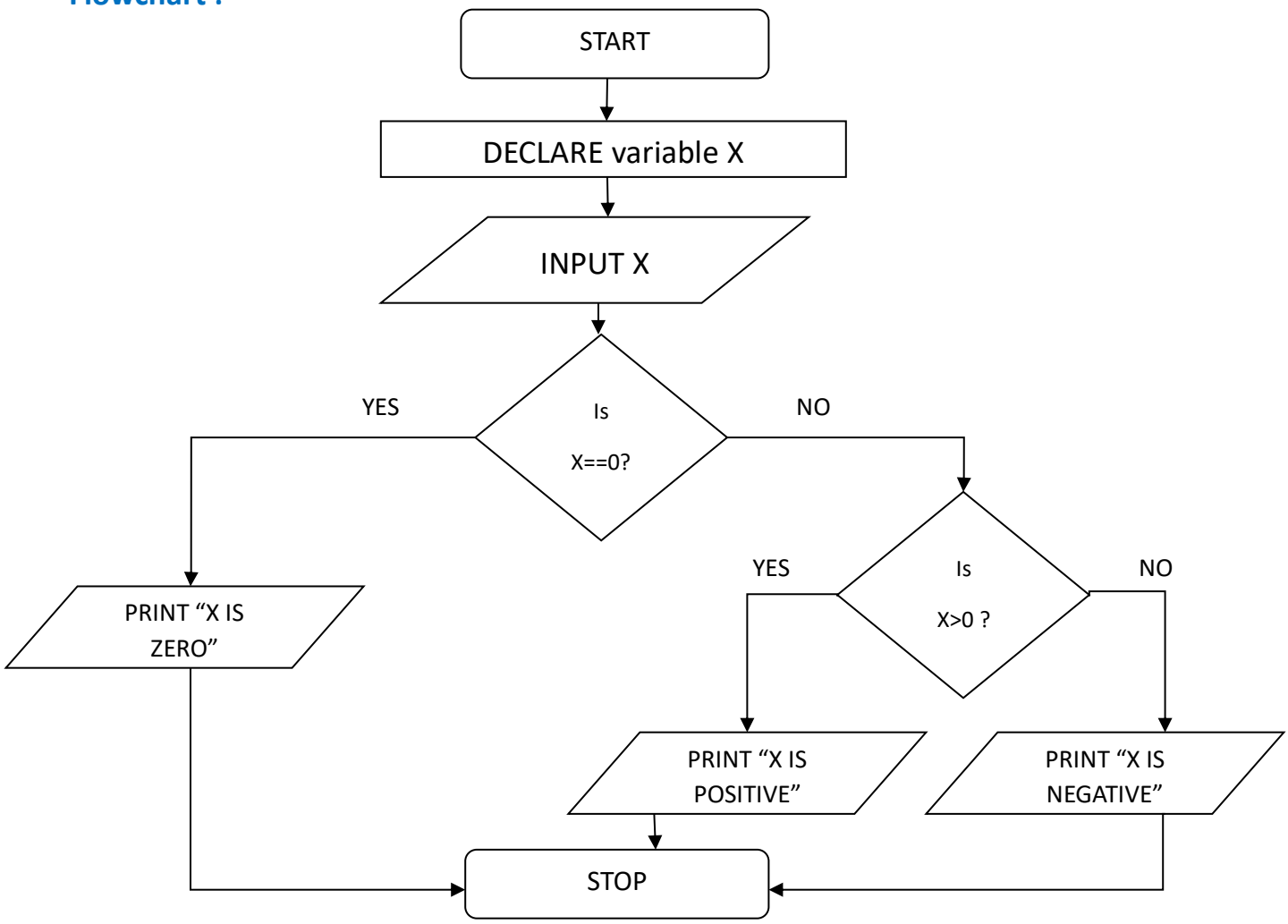
PRINT "Number is Positive"

Step 6 : ELSE

PRINT "Number is Negative"

Step 7 : STOP

Flowchart :


START
DECLARE variable X
INPUT X
Is
X==0?
YES
NO
PRINT “X IS
ZERO”
Is
X>0 ?
YES
NO
PRINT “X IS
POSITIVE”
PRINT “X IS
NEGATIVE”
STOP

Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int x;
   clrscr();
   printf("\nEnter Number :");
   scanf("%d",&x);
   if(x==0){
      printf("Number %d, is zero", x);
   }
   else if(x>0){
      printf("Number %d, is positive", x);
   }
   else
      printf("Number %d, is negative", x);

   getch();
}
```

Output :

